Dai *et al. Cybersecurity* (2025) 8:105
https://doi.org/10.1186/s42400-025-00439-5

Cybersecurity

RESEARCH

Open Access

Lattice-based, LWE-leakage model for Gaussian and uniform secret and its application in decentralization

Xiaokang Dai[1,2], Jingwei Chen[1,2], Wenyuan Wu[1,2*] and Yong Feng[1,2]

Abstract

In the case of standard LWE samples $(\mathbf{A}, \mathbf{b} = \mathbf{sA} + \mathbf{e})$, $\mathbf{A}$ is typically uniformly over $\mathbb{Z}_q^{n\times m}$. Under the DLWE assumption, the conditional distribution of $\mathbf{s}|(\mathbf{A}, \mathbf{b})$ and $\mathbf{s}$ is expected to be consistent. However, in the case where an adversary chooses $\mathbf{A}$ adaptively, the disparity between the two entities may be larger. In this work, our primary focus is on the quantification of the Average Conditional Min-Entropy $\tilde{H}_\infty(\mathbf{s}|\mathbf{sA} + \mathbf{e})$ of $\mathbf{s}$, where $\mathbf{A}$ is chosen by the adversary. Brakerski and Döttling answered the question in one case: they proved that when $\mathbf{s}$ is uniformly chosen from $\mathbb{Z}_q^n$, it holds that $\tilde{H}_\infty(\mathbf{s}|\mathbf{sA} + \mathbf{e}) \propto \rho_\sigma(\Lambda_q(\mathbf{A}))$. We prove that for any $d \leq q$, when $\mathbf{s}$ is uniformly chosen from $\mathbb{Z}_d^n$ or is sampled from a discrete Gaussian distribution, there are also similar results. As an application of the above results, we improved the multi-key fully homomorphic encryption and answered the question raised at the end of their work positively: we have GSW-type ciphertext rather than Dual-GSW, and the improved scheme has shorter keys and ciphertexts.

Keywords Leftover hash lemma, Leakage resilient cryptography, Multi-key FHE

Introduction

Secure multi-party computation (MPC) (Yao 1982), Threshold fully homomorphic encryption (ThFHE) and Multi-key fully homomorphic encryption (MKFHE) (Lopez-Alt et al. 2012) provide technical support for computing tasks involving multiple users. Depending on the assumptions, the techniques mentioned above can be divided into two categories: the first with setup (trusted third party, common reference string (CRS)), while the second without setup (plain model).

Compared to schemes or protocols under the plain model, those schemes that involve a trusted third party or CRS are much simpler and more efficient, particularly during the initialization phase. However, some people believe that introducing such assumptions seems like cheating (since there is such a trusted third party, why not put everyone's data in his hands and then return the results to all parties.) Therefore, building cryptographic primitives under the plain model has also become a demand for some people.

The key issue here is that the initialization of MPC, ThFHE, or MKFHE protocols, such as key generation, often relies on some common parameters. If these parameters come from a trusted third party, their integrity can be guaranteed. If there is no trusted third party

This work is the extend version of conference paper.

*Correspondence:
Wenyuan Wu
wuwenyuan@cigit.ac.cn
[1] Chongqing Key Laboratory of Automated Reasoning and Cognition, Chongqing Institute of Green and Intelligent Technology of Chinese Academy of Sciences, Chongqing, China
[2] Chongqing School, University of Chinese Academy of Sciences, Chongqing, China

Springer Open

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

or CRS, then the initialization of the protocol is usually an interactive process involving users. At this time, the reliability of the data cannot be guaranteed, which may result in the compromise of user privacy. For example, in the MKFHE scheme (Brakerski et al. 2017), parties need to multiply their own private key $\mathbf{s}$ with a matrix $\mathbf{A}$ generated by another party and make $\mathbf{sA}$ public in order to support *ciphertext expansion*. In the oblivious transfer protocol (Brakerski and Döttling 2018), the first round message $\mathbf{y} = \mathbf{tA} + \mathbf{e}$ of the sender is composed of its own secret $\mathbf{t}$ multiplied by $\mathbf{A}$ generated by the receiver plus a small error. Similarly, the unbounded MPC protocol (Ananth et al. 2021) also requires the LWE samples $\mathbf{y} = \mathbf{sA} + \mathbf{e}$ to be made public, where $\mathbf{A}$ is generated by the adversary.

Motivation

In the MKFHE scheme (Brakerski et al. 2017), assuming there are k parties, in order to support subsequent *ciphertext expansion*, each party needs to multiply their own private key $\mathbf{s}$ by the public keys $\{\mathbf{A}_i\}_{i\in[k-1]}$ of the other $k-1$ parties and make $\{\mathbf{b}_i = \mathbf{sA}_i\}_{i\in[k-1]}$ public. To quantify the average conditional min-entropy $\tilde{H}_\infty(\mathbf{s}|\{\mathbf{b}_i\}_{i\in[k-1]})$ of $\mathbf{s} \in \{0,1\}^m$ after disclosing $\{\mathbf{b}_i = \mathbf{sA}_i\}_{k-1}$, we estimate the leakage in the worst case. For $\mathbf{b}_i \in \mathbb{Z}_q^n$, $\{\mathbf{b}_i = \mathbf{sA}_i\}_{i\in[k-1]}$ leaks $\mathbf{s}$ with a maximum of $(k-1)n\log q$ bits. According to the proof in Brakerski et al. (2017), based on the leftover hash lemma (LHL), in order to ensure that the statistical distance between the ciphertext and the uniform distribution is less than $2^{-\lambda}$, m should at least satisfy $m-(k-1)n\log q \geq \log q + 2\lambda$. In Brakerski and Döttling (2018), another "active leakage" model $\mathbf{s}|\mathbf{b} = \mathbf{sA} + \mathbf{e}$ was applied, to ensure that the entropy of $\mathbf{s}$ remains sufficient after $\mathbf{b} = \mathbf{sA} + \mathbf{e}$ is disclosed, it proved that $\tilde{H}_\infty(\mathbf{s}|\mathbf{sA}+\mathbf{e}) \geq -\log(1/\rho_\sigma(\Lambda_q(\mathbf{A})) + 2^{-m})$. We believe that the model $\mathbf{s}|\mathbf{sA}+\mathbf{e}$ is a better "active leakage" model compared to $\mathbf{s}|\mathbf{sA}$, because $\tilde{H}_\infty(\mathbf{s}|\mathbf{sA}+\mathbf{e})$ establishes a relationship with $\Lambda_q(\mathbf{A})$. Additionally, provided that $\Lambda_q(\mathbf{A})$ has enough short vectors the loss ratio of former is $O(1/\log q)$, whereas the latter is $O(1/n)$. Based on this, the work (Brakerski and Döttling 2018) constructed the first post-quantum secure oblivious transfer protocol under the plain model that can resist malicious receivers. So far, we have seen two "active leakage" models: $\mathbf{s}|\mathbf{sA}$ and $\mathbf{s}|\mathbf{sA}+\mathbf{e}$. $\tilde{H}_\infty(\mathbf{s}|\mathbf{sA})$ quantifies the conditional entropy of $\mathbf{s}|\mathbf{sA}$ in a more rudimentary way, while $\tilde{H}_\infty(\mathbf{s}|\mathbf{sA}+\mathbf{e})$ quantifies the conditional entropy of $\mathbf{s}|\mathbf{sA}+\mathbf{e}$ based on the properties of lattices, but is limited to $\mathbf{s} \leftarrow \mathbb{Z}_q^n$. We are interested in whether there is a similar result for any $d \leq q$, where $\mathbf{s} \leftarrow \mathbb{Z}_d^n$, or $\mathbf{s}$ is sampled from a discrete Gaussian distribution.

Such a requirement is not baseless. In the LWE-like sample $\mathbf{sA} + \mathbf{e}$, it is sometimes convenient and necessary to bound the norm of $\mathbf{s}$. In order to support bootstrapping in FHE, it is necessary to encrypt the private key $\mathbf{s}$. If $\mathbf{s}$ is uniformly distributed over $\mathbb{Z}_q$, how can it be filled into the plaintext space? One method is to bit-decomposition the key before encrypting it. For example, the work (Ducas and Micciancio 2015; Chillotti et al. 2016) adopt binary key to fill into the plaintext space in order to realize bootstrapping more quickly and control the growth of noise. Another approach is to limit the norm of the $\mathbf{s}$ to a small range. Therefore, Applebaum et al. (2009) reduced the LWE samples with discrete Gaussian secrets to the LWE samples with uniform secrets. MKFHE scheme (Chen et al. 2019) requires that $\mathbf{s}$ be sampled from the discrete Gaussian distribution in order to mitigate the noise introduced by the *re-linearization* after multiplication of the ciphertext. In addition, the work (Ananth et al. 2021) utilizes the result of Brakerski and Döttling (2018) to defend against semi-malicious adversaries, however, in their proposed scheme, $\mathbf{s}$ is drawn from discrete Gaussian distribution.

Therefore, if we can quantify $\tilde{H}_\infty(\mathbf{s}|\mathbf{sA}+\mathbf{e})$ where $\mathbf{s} \leftarrow \mathbb{Z}_d^n$ (for any $d \leq q$) or $\mathbf{s}$ is taken from a discrete Gaussian distribution, we believe that this result can be applied in many ways. Specifically, based on this result, we optimized the MKFHE (Brakerski et al. 2017), resulting in shorter keys and smaller ciphertexts. We present our results in the following section.

Our results

For LWE samples whose secrets are sampled from discrete Gaussian distribution, we have the following result.

Theorem 1 *Let n, q, $m = O(n\log q)$ be integers, and $0 < \sigma < q/(2\sqrt{m+n})$. For the given matrix $\tilde{\mathbf{A}} = (\bar{\mathbf{A}}, \mathbf{A}') \in \mathbb{Z}_q^{n\times n} \times \mathbb{Z}_q^{n\times m}$, where $\bar{\mathbf{A}}$ is invertible, let $\mathbf{A} = -\bar{\mathbf{A}}^{-1}\mathbf{A}'$, $\mathbf{s} \leftarrow \mathcal{D}_{\mathbb{Z}^n,\sigma}$, $\mathbf{e} \leftarrow \mathcal{D}_{\mathbb{Z}^m,\sigma}$. It holds that*

$$\tilde{H}_\infty(\mathbf{s}|\mathbf{sA}+\mathbf{e}) \geq -\log\left(\frac{1}{\rho_\sigma(\Lambda_q(\tilde{\mathbf{A}}))} + 2^{-(m+n)}\right).$$

In practical applications, we need to estimate a specific value of $\tilde{H}_\infty(\mathbf{s}|\mathbf{sA}+\mathbf{e})$. According to the result of Brakerski and Döttling (2018), assuming that $\Lambda_q(\tilde{\mathbf{A}})$ contains at least $n/2$ short vectors, we have the infimum $\tilde{H}_\infty(\mathbf{s}|\mathbf{sA}+\mathbf{e}) \geq n$.

For the LWE samples whose secrets are sampled from bounded uniform distribution, we present a more general results of Lemma 3.2 in Brakerski and Döttling (2018) (Lemma 3.2 is a special case of our Theorem 2).

Theorem 2 *Let n, d, q, $m = O(n \log d)$ be integers and $0 < d \leq q$, $0 < \sigma < d/\sqrt{m}$. For the given $\mathbf{A} \in \frac{q}{d}\mathbb{Z}_d^{n \times m}$, let $\mathbf{s} \leftarrow \mathbb{Z}_d^n$ and $\mathbf{e} \leftarrow \mathcal{D}_{\frac{q}{d}\mathbb{Z}^m,\sigma}$, then it holds that*

$$\tilde{H}_\infty(\mathbf{s}|\mathbf{sA}+\mathbf{e}) \geq -\log\left(\frac{1}{\rho_\sigma(\Lambda_q(\mathbf{A}))} + 2^{-m}\right).$$

Clearly, when $d = q$ Theorem 2 degenerates to Lemma 3.2 in Brakerski and Döttling (2018).

New results (compared with the conference version (Dai et al. 2024))

In our previous conference paper, we quantified the lower bounds of the conditional entropy of private key $\mathbf{s}$ when it comes from a discrete Gaussian distribution or a bounded uniform distribution, respectively as in Theorem 1 and Theorem 2. Furthermore, we construct a MKFHE scheme on $\frac{q}{d}\mathbb{Z}^m$. This is because for $\mathbf{s} \leftarrow \mathbb{Z}_d^n$, we can quantify the lower bound of the conditional entropy of $\mathbf{s}$ as in Theorem 2, and we can also prove the leakage-resilient property of the leftover hash lemma on the uniform distribution as in Theorem 5. However, when the private key $\mathbf{s}$ is taken from a discrete Gaussian distribution, although we can quantified the lower bound of the conditional entropy of $\mathbf{s}$ as in Theorem 1, we cannot prove the leakage-resilient property of the leftover hash lemma on the discrete Gaussian distribution. This is why we construct MKFHE on $\frac{q}{d}\mathbb{Z}^m$. Recently, Jin et al. (2024) optimized the leftover hash lemma over discrete Gaussian, reducing the variance by a factor of $\omega(\sqrt{\log \lambda})$. In addition, they proved the leftover hash lemma over discrete Gaussian is also leak-resistant, similar to the general leftover hash lemma. Based on Theorem 1 and the result of Jin et al. (Theorem 5.1 in Jin et al. (2024)), we construct the MKFHE *Scheme*#1 on $\mathbb{Z}^m$. Compared to our previous scheme constructed on $\frac{q}{d}\mathbb{Z}^m$ and the MKFHE in Brakerski et al. (2017), *Scheme*#1 is more intuitive and follows standard conventions and has smaller key and ciphertext.

Based on Theorem 2 and the regularity of universal hash functions mapped to prime order groups (we give a proof in C), we construct *Scheme*#2. Compared with scheme (Brakerski et al. 2017), the private key is taken from bounded uniform distribution, and it also has smaller ciphertext and keys. In particular, we provide a positive answer to the question raised at the end of Brakerski et al. (2017): the ciphertext of *Scheme*#1 and *Scheme*#2 are GSW-like manner, rather than Dual GSW. The comparison as shown in Table 1.

It is worth noting that in Brakerski et al. (2017), when computing the auxiliary key $\{\mathbf{b}_i = \mathbf{sA}_i\}_{i\in[k]}$, each participant needs to multiply their own secret key $\mathbf{s}$ with the public key matrix $\{\mathbf{A}_i\}_{i\in[k]}$ of other participants, and then publish $\{\mathbf{b}_i\}_{i\in[k]}$. Since $\{\mathbf{A}_i\}_{i\in[k]}$ are generated by a semi-malicious adversary, this process may lead to leakage of the secret key $\mathbf{s}$. To ensure that the conditional entropy of $\mathbf{s}$ remains sufficiently high after publishing $\{\mathbf{b}_i\}_{i\in[k]}$, the length of $\mathbf{s}$ must be carefully chosen based on the number of participants. In contrast, our scheme constructs the auxiliary key in an LWE form as $\mathbf{b} = \mathbf{s}(\mathbf{A}_1, \mathbf{A}_2, \cdots, \mathbf{A}_k) + (\mathbf{e}_1, \mathbf{e}_2, \cdots \mathbf{e}_k)$. To ensure that the conditional entropy of $\mathbf{s}$ is large enough, we require the q-ary lattice generated by the matrix $\mathbf{A} = (\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_k)$ to contain at least $n/2$ short vectors, instead of increasing the length of $\mathbf{s}$ as the number of participants grows.

Related works

The work of Brakerski and Döttling (2020a) on the hardness of LWE on general entropic distributions was dedicated to proving the hardness of entropy LWE: for a key distribution $\mathcal{S}$ with support over $\mathbb{Z}^n$, assuming that $\tilde{H}_\infty(\mathbf{s}|\mathbf{s}+\mathbf{e})$ is large enough, then the entropy LWE is hard (equivalent to the generalization of Goldwasser et al's work (2010), which proved that when the key $\mathbf{s}$ is taken from $\{0,1\}$, and $\tilde{H}_\infty(\mathbf{s})$ is large enough, the binary LWE is leakage-resilient). We must point out that our work is dedicated to characterizing the lower bound of $\tilde{H}_\infty(\mathbf{s}|\mathbf{sA}+\mathbf{e})$, where $\mathbf{A}$ may not be uniformly distributed. This type of leakage model is more prevalent in multi-party cooperation protocols, such as oblivious transfer or MKFHE. The leakage model of $\tilde{H}_\infty(\mathbf{s}|\mathbf{s}+\mathbf{e})$ is more in line with the side channel attack (in our work, it becomes passive leakage). In contrast to our theoretical

Table 1 Complexity

Scheme	Ciphertext size	Hom-multiplication	Communication in setup	Structure	Setup
Brakerski et al. (2017)	$O(k^2n^2\log^4 q)$	$O(k^6n^3\log^5 q)$	$O(kn^2\log^2 q)$	Dual-GSW	–
Scheme#1	$O(n^2\log^2 q)$	$O(k^3n^3\log^2 q)$	$O(n^2\log^2 q)$	GSW	–
Scheme#2	$O(n^2\log^2 d)$	$O(k^3n^3\log^2 d)$	$O(n^2\log^2 d)$	GSW	–

tk, n, q, d denotes number of parties, LWE dimension, modulus respectively and $d = q/\mathrm{poly}(\lambda)$. The ciphertext are counted in bits. The Hom-multiplication column counts the number of multiplications on $\mathbb{Z}_q$ required for a homomorphic multiplication. The Communication in setup column counts the communication traffic required for the interactive key generation phase. The Structure column indicates whether the scheme has a GSW ciphertext structure or a dual-GSW ciphertext structure. The Setup column indicates whether the scheme is initialized with CRS, "–" means no

quantification of the conditional min-entropy under the LWE setting, Xu et al. (2025) focused on practical parameter optimization of LWE-based encryption schemes using a monotonic optimization framework. Their methodology provides an automatic way to select parameters such as , , to balance security and performance, which is complementary to our entropy-based analysis.

Therefore, we believe that these two works should complement each other. Their research focuses on the hardness of entropy LWE, and considers how to quantify $\tilde{H}_\infty(\mathbf{s}|\mathbf{s}+\mathbf{e})$, which provides more confidence for leakage-resilient cryptography. However, our work focuses on characterizing the active leakage of $\mathbf{s}|\mathbf{sA}+\mathbf{e}$ (there should be no side channel to obtain $\mathbf{s}$ from $\mathbf{sA}+\mathbf{e}$), which provides a tool for further weakening the setup (without CRS, trusted third party) in the MPC and MKFHE.

Road-map

In Sect. "Preliminaries" we define some notation and introduce fundamental definition and results on lattice, establishing the groundwork for subsequent discussions. In Sect. "Technical overview" we provide a technical overview of our approach, emphasizing the ideas behind the model and providing intuition for the results, followed by detailed proofs in later sections. In Sect. "Lattice-based, more general leakage-resilient model" we proved a more general result for the Average Conditional Min-Entropy $\tilde{H}_\infty(\mathbf{s}|\mathbf{sA}+\mathbf{e})$, extending previous work and quantifying the leakage-resilient properties in lattice-based settings. In Sect. "The hardness of entropy LWE" we prove the hardness of entropy LWE, where the key $\mathbf{s}$ is taken from a discrete Gaussian distribution. This result will be used in the subsequent security proof of *Scheme*#1. In Sect. "Optimized multi-key fully homomorphic encryption scheme" we discusses the construction and improvement of MKFHE, with a detailed analysis of *Scheme*#1, showcasing its security under the plain model.

Preliminaries

Notation

Let $\mathsf{negl}(\lambda)$ be a negligible function parameterized by λ. Lowercase bold letters such as $\mathbf{v}$, unless otherwise specified, represent vectors. Vectors are typically represented as row vectors, while matrices are denoted by uppercase bold letters such as $\mathbf{M}$. Let k be an integer, $[k]$ be the set of integers $\{1,\cdots,k\}$. If X is a distribution, then $a \leftarrow X$ denotes that the value a is chosen according to the distribution X. If X is a finite set, then $a \leftarrow X$ denotes that the value of a is uniformly sampled from X. For two distributions X and Y, we use $X \approx_s Y$ to represent that X and Y are statistically indistinguishable, while $X \approx_c Y$ represents that they are computationally indistinguishable.

To decompose elements in $\mathbb{Z}_q$ into binary, we review the Gadget matrix (Micciancio and Peikert 2012; Alperin-Sheriff and Peikert 2014) here. Let $\mathbf{G}^{-1}(\cdot)$ be the computable function that for any $\mathbf{M} \in \mathbb{Z}_q^{m\times n}$, it holds that $\mathbf{G}^{-1}(\mathbf{M}) \in \{0,1\}^{ml\times n}$, where $l = \lceil \log q \rceil$. Let $\mathbf{g} = (1,2,\ldots,2^{l-1}) \in \mathbb{Z}_q^l$, $\mathbf{G} = \mathbf{I}_m \otimes \mathbf{g} \in \mathbb{Z}_q^{m\times ml}$, it satisfies $\mathbf{GG}^{-1}(\mathbf{M}) = \mathbf{M}$.

Gadget decomposition over $\frac{q}{d}\mathbb{Z}_d$

As *Scheme*#2 is constructed on $\frac{q}{d}\mathbb{Z}^m$, here we consider the bit decomposition on $\frac{q}{d}\mathbb{Z}_d$ which is essentially the same as the bit decomposition on $\mathbb{Z}_q$ but multiplied by q/d. Let d, q be two integers and $d \le q$, $\mathbf{g} = \frac{q}{d}(1,2,\ldots,2^{l-1})$ where $l = \lceil \log d \rceil$, for any $a \in \frac{q}{d}\mathbb{Z}_d$ let $a = \frac{q}{d}\cdot t$ where $t \in \mathbb{Z}_d$, we define $\mathbf{g}^{-1}(a) = \{0,1\}^l$ as the decomposition of t, it holds that $\mathbf{g}\cdot\mathbf{g}^{-1}(a) = a$. Furthermore, for $\mathbf{M} \in \frac{q}{d}\mathbb{Z}_d^{m\times n}$ let $\mathbf{G} = \mathbf{I}_m \otimes \mathbf{g}$ it holds that $\mathbf{G}^{-1}(\mathbf{M}) \in \{0,1\}^{ml\times n}$ and $\mathbf{GG}^{-1}(\mathbf{M}) = \mathbf{M}$.

Semi-malicious adversary (in Asharov et al. (2012))

The semi-malicious adversary model was first introduced by Asharov et al. in their 2012 work (Asharov et al. 2012). The motivation behind this model is that the semi-honest adversary assumption is often too weak, as real-world adversaries rarely follow the protocol specification exactly. The semi-malicious model captures an adversary whose power lies between that of semi-honest and fully malicious adversaries. Formally, it is modeled as an interactive Turing machine that, in addition to the standard tapes, has access to a special *witness tape*. Whenever the adversary outputs a protocol message m, it must also write on the witness tape a description of how m was generated. This requirement restricts the adversary's capabilities to some extent, particularly when m is generated through a trapdoor mechanism. For instance, suppose the protocol specifies that $m = \langle \mathbf{s}, \mathbf{a}\rangle \mod q$, where $\mathbf{a}$ is uniformly sampled from $\mathbb{Z}_q^n$ and $\mathbf{s} \in \{0,1\}^n$. If the adversary wants to output a specific m', it must find some binary vector $\mathbf{s}' \in \{0,1\}^n$ such that $m' = \langle \mathbf{s}', \mathbf{a}\rangle \mod q$, and then write $\mathbf{s}'$ to the witness tape. In other words, writing to the witness tape is equivalent to solving a small integer solution (SIS) problem. This example demonstrates how the witness tape can limit the adversary's behavior.

On the other hand, there are situations where the witness tape fails to constrain the adversary. For example, if the protocol specifies that m is a uniformly random matrix over $\mathbb{Z}_q^{n\times m}$, then the adversary can arbitrarily choose a matrix $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$, write it to the witness tape, and claim it was sampled uniformly at random. This shows a limitation of the semi-malicious model in unstructured scenarios.

Average conditional min-entropy (in Brakerski and Döttling (2020a))

Let X be a random-variable supported on a finite set $\mathcal{X}$, and let Z be a random variable supported on a finite set $\mathcal{Z}$. The average-conditional min-entropy $\tilde{H}_\infty(X|Z)$ of X given Z is defined as

$$\tilde{H}_\infty(X|Z) = -\log\left(E_z\left[\max_{x\in\mathcal{X}} \Pr[X = x|Z = z]\right]\right).$$

Some result on the lattice

Theorem 3 *Let Λ be a lattice, V be the Voronoï cell of Λ, $\mathbf{t}, \mathbf{t}'$ are two vectors in span(Λ), then the following three statements are equivalent*

1. $\mathbf{t}'$ *is the shortest vector in* $\mathbf{t} + \Lambda$.
2. $\mathbf{t}' \in (\mathbf{t} + \Lambda) \bigcap V$.
3. $\mathbf{v} = \mathbf{t} - \mathbf{t}' \in \Lambda$ *is the nearest lattice point to* $\mathbf{t}$.

Definition 1 Let $\rho_\sigma(\mathbf{x}) = \exp(-\pi||\mathbf{x}/\sigma||^2)$ be a Gaussian function scaled by a factor of $\sigma > 0$. Let $\Lambda \subset \mathbb{R}^m$ be a lattice, and $\mathbf{c} \in \mathbb{R}^m$. The discrete Gaussian distribution $\mathcal{D}_{\Lambda+\mathbf{c},\sigma}$ with support $\Lambda + \mathbf{c}$ is defined as

$$\mathcal{D}_{\Lambda+\mathbf{c},\sigma}(\mathbf{x}) = \frac{\rho_\sigma(\mathbf{x})}{\rho_\sigma(\Lambda + \mathbf{c})}.$$

Lemma 1

(*Lemma* 3.1 *in Brakerski and Döttling* (2018)) *Let $\Lambda \subseteq \Lambda_0 \subseteq \mathbb{R}^m$ be full rank lattices and let $T \subseteq \Lambda_0$ be a system of coset representatives of Λ_0/Λ, i,e. we can write every $\mathbf{x} \in \Lambda_0$ as $\mathbf{x} = \mathbf{t} + \mathbf{z}$ for unique $\mathbf{t} \in \Lambda$ and $\mathbf{z} \in \Lambda$. Then it holds for any parameter $\sigma > 0$ that*

$$\frac{\rho_\sigma(T)}{\rho_\sigma(\Lambda_0)} \le \frac{1}{\rho_\sigma(\Lambda)}.$$

Lemma 2

(*Lemma* 4.3 *in Brakerski and Döttling* (2018)) *Let $\mathcal{B}$ be the unit ball. Let $\Lambda \in \mathbb{R}^m$, $\sigma > 0$ and $\gamma > 0$ be such that $\Lambda \bigcap \gamma\mathcal{B}$ contains at least k linearly independent vectors. Then it holds that $\rho_\sigma(\Lambda) \ge (\sigma/\gamma)^k$.*

Lemma 3

(*Lemma* 1.5 *in Banaszczyk* (1993)) *Let $\sigma > 0$ and $\mathbf{x} \leftarrow \mathcal{D}_{\mathbb{Z}^m,\sigma}$. Then it holds that $||\mathbf{x}|| \le \sigma \cdot \sqrt{m}$, except with probability 2^{-m}.*

Theorem 4 (adapted from Theorem 5.3 in Jin et al. (2024)) *Let $q, n, r \ge 1$ and $n > r$ be integers with q prime. Let $\mathcal{D}_{\mathbb{Z}^n,\sigma}$ be the discrete Gaussian distribution defined over $\mathbb{Z}^n$ with standard deviation σ. Let $\mathbf{s} \leftarrow \mathcal{D}_{\mathbb{Z}^n,\sigma}$ and $\mathsf{aux}(\mathbf{s})$ is the auxiliary information on $\mathbf{s}$. Let $\mathcal{D}_0 = (U(\mathbb{Z}_q^{n\times r} \times \mathbb{Z}_q^r), \mathsf{aux}(\mathbf{s}))$, $\mathcal{D}_1 = (\mathbf{A}, \mathbf{sA}, \mathsf{aux}(\mathbf{s}))$ where $\mathbf{A} \leftarrow \mathbb{Z}_q^{n\times r}$. For any positive $\epsilon < 2^{(\delta-1)/2}$ and $\delta > 0$, if $H_\infty(\mathbf{s}|\mathsf{aux}(\mathbf{s})) \ge n\log\sigma - \delta$ and $n\log\sigma \ge 2\log(1/\epsilon) + r\log q + \delta$, it holds that $\Delta(\mathcal{D}_0, \mathcal{D}_1) \le \epsilon$.*

Learning with errors

The Learning With Errors (LWE) problem was introduced by Regev (2009). In general, we are primarily interested in its decision version.

Definition 2 (Decision-LWE) For $n, m, q \in \mathbb{N}$ and real $\sigma > 0$, the $\mathsf{DLWE}_{n,m,q,\sigma}$ problem is to distinguish the following distribution

- $\mathcal{D}_0$: the jointly distribution $(\mathbf{A}, \mathbf{z}) \in (\mathbb{Z}_q^{n\times m} \times \mathbb{Z}_q^m)$ is sampled by $\mathbf{A} \leftarrow \mathbb{Z}_q^{n\times m}, \mathbf{z} \leftarrow \mathbb{Z}_q^m$.
- $\mathcal{D}_1$: the jointly distribution $(\mathbf{A}, \mathbf{b} = \mathbf{sA} + \mathbf{e}) \in (\mathbb{Z}_q^{n\times m} \times \mathbb{Z}_q^m)$, where $\mathbf{A} \leftarrow \mathbb{Z}_q^{n\times m}, \mathbf{s} \leftarrow \mathbb{Z}_q^n, \mathbf{e} \leftarrow \mathcal{D}_{\mathbb{Z}^m,\sigma}$.

As shown in Regev (2009), for certain module q and $\sigma = \alpha q \ge 2\sqrt{n}$, the $\mathsf{DLWE}_{n,m,q,\sigma}$ is true as long as certain worst-case lattice problem is hard to solve using a quantum algorithm. The matrix version of this problem ask to distinguish $(\mathbf{A}, \mathbf{SA} + \mathbf{E})$ from $(\mathbf{A}, \mathbf{U})$ where $\mathbf{S} \leftarrow \mathbb{Z}_q^{k\times n}$, $\mathbf{E} \leftarrow \mathcal{D}_{\mathbb{Z}^{k\times m},\sigma}$ and $\mathbf{U} \leftarrow \mathbb{Z}_q^{k\times m}$, whose hardness for any $k = \mathsf{poly}(n)$ can be established from $\mathsf{DLWE}_{n,m,q,\chi}$ via a routine hybrid-argument.

Technical overview

In this section, we will briefly outline our technical approach, focusing on our ideas and providing readers with some intuition. A detailed description will be given in the subsequent section. For the given $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$, and $\mathbf{y} = \mathbf{sA} + \mathbf{e}$, where $\mathbf{s} \leftarrow \mathbb{Z}_q^n$, $\mathbf{e} \leftarrow \mathcal{D}_{\mathbb{Z}^m,\sigma}$, by the definition of Average Conditional Min-Entropy 2.1.3

$$\tilde{H}_\infty(\mathbf{s}|\mathbf{y}=\mathbf{sA}+\mathbf{e}) = -\log\left(\mathsf{E}_\mathbf{y}\left[\max_{\mathbf{s}^*}\Pr_{\mathbf{s},\mathbf{e}}[\mathbf{s}=\mathbf{s}^*|\mathbf{y}=\mathbf{sA}+\mathbf{e}]\right]\right).$$

where $\mathbf{s}^*$ is the point maximizes the conditional probability $\Pr_{\mathbf{s},\mathbf{e}}[\mathbf{s}=\mathbf{s}^*|\mathbf{y}=\mathbf{sA}+\mathbf{e}]$. The work (Brakerski and Döttling 2018) notes that when $\mathbf{s}$ is uniformly chosen from $\mathbb{Z}_q^n$, by Bayes' rule,

$$\begin{aligned}\Pr_{\mathbf{s},\mathbf{e}}[\mathbf{s}=\mathbf{s}^*|\mathbf{y}=\mathbf{sA}+\mathbf{e}] &= \Pr_{\mathbf{s},\mathbf{e}}[\mathbf{y}=\mathbf{sA}+\mathbf{e}\mid\mathbf{s}=\mathbf{s}^*]\cdot\frac{\Pr[\mathbf{s}=\mathbf{s}^*]}{\Pr_{\mathbf{s},\mathbf{e}}[\mathbf{y}=\mathbf{sA}+\mathbf{e}]}\\ &= \Pr_{\mathbf{e}}[\mathbf{e}=\mathbf{y}-\mathbf{s}^*\mathbf{A}]\\ &\cdot\frac{\Pr[\mathbf{s}=\mathbf{s}^*]}{\sum_{\mathbf{s}'}\Pr_{\mathbf{e}}[\mathbf{y}=\mathbf{sA}+\mathbf{e}\mid\mathbf{s}=\mathbf{s}']\Pr[\mathbf{s}=\mathbf{s}']}\\ &= \Pr_{\mathbf{e}}[\mathbf{e}=\mathbf{y}-\mathbf{s}^*\mathbf{A}]\cdot\frac{q^{-n}}{\sum_{\mathbf{s}'}\Pr_{\mathbf{e}}[\mathbf{e}=\mathbf{y}-\mathbf{s}'\mathbf{A}]\cdot q^{-n}}\\ &= \frac{\Pr_{\mathbf{e}}[\mathbf{e}=\mathbf{y}-\mathbf{s}^*\mathbf{A}]}{\sum_{\mathbf{s}'}\Pr_{\mathbf{e}}[\mathbf{e}=\mathbf{y}-\mathbf{s}'\mathbf{A}]}.\end{aligned}$$

The denominator is a constant, that is, when $\mathbf{s}^*\mathbf{A}$ is the point closest to $\mathbf{y}$, conditional probability $\Pr_{\mathbf{s},\mathbf{e}}[\mathbf{s}=\mathbf{s}^*|\mathbf{y}=\mathbf{sA}+\mathbf{e}]$ is the largest. The events that $\mathbf{s}^*\mathbf{A}$ is the lattice point closest to $\mathbf{y}$ and $\mathbf{e}\in V$, are equivalent, where V is the discrete *Voronoïcell* of $\mathbf{s}^*\mathbf{A}$. Therefore, they can transform the problem from finding the probability $\Pr_{\mathbf{s},\mathbf{e}}[\mathbf{s}=\mathbf{s}^*|\mathbf{y}=\mathbf{sA}+\mathbf{e}]$ to finding the probability $\Pr(\mathbf{e}\in V)$.

LWE with discrete Gaussian secrets

However, when $\mathbf{s}$ is sampled from a discrete Gaussian distribution, we cannot directly apply the above method to quantify the probability $\Pr_{\mathbf{s},\mathbf{e}}[\mathbf{s}=\mathbf{s}^*|\mathbf{y}=\mathbf{sA}+\mathbf{e}]$. The reason is as follows, also according to Bayes' rule

$$\begin{aligned}\Pr_{\mathbf{s},\mathbf{e}}[\mathbf{s}=\mathbf{s}^*|\mathbf{y}=\mathbf{sA}+\mathbf{e}] &= \Pr_{\mathbf{s},\mathbf{e}}[\mathbf{y}=\mathbf{sA}+\mathbf{e}\mid\mathbf{s}=\mathbf{s}^*]\\ &\cdot\frac{\Pr[\mathbf{s}=\mathbf{s}^*]}{\Pr_{\mathbf{s},\mathbf{e}}[\mathbf{y}=\mathbf{sA}+\mathbf{e}]} = \Pr_{\mathbf{s},\mathbf{e}}[\mathbf{e}=\mathbf{y}-\mathbf{s}^*\mathbf{A}]\\ &\cdot\frac{\Pr[\mathbf{s}=\mathbf{s}^*]}{\sum_{\mathbf{s}'}\Pr_{\mathbf{s},\mathbf{e}}[\mathbf{y}=\mathbf{sA}+\mathbf{e}\mid\mathbf{s}=\mathbf{s}']\Pr[\mathbf{s}=\mathbf{s}']}.\end{aligned}$$

We cannot proceed to the next step because, when $\mathbf{s}$ is drawn from a discrete Gaussian distribution, we cannot determine the probability that $\mathbf{s}$ equals $\mathbf{s}^*$. At this time, the point $\mathbf{s}^*\mathbf{A}$ that maximizes the probability $\Pr_{\mathbf{s},\mathbf{e}}[\mathbf{s}=\mathbf{s}^*|\mathbf{y}=\mathbf{sA}+\mathbf{e}]$ is not necessarily the lattice point closest to $\mathbf{y}$. This is the challenge of determining the probability $\Pr_{\mathbf{s},\mathbf{e}}[\mathbf{s}=\mathbf{s}^*|\mathbf{y}=\mathbf{sA}+\mathbf{e}]$ when $\mathbf{s}$ is drawn from a discrete Gaussian distribution.

By the reduction from the LWE with uniform secrets to the LWE with Gaussian secrets, the noise of the former becomes the secrets of the latter. Therefore, in order to quantify the entropy of the latter secrets, we can turn to the entropy of the noise. By definition

$$\tilde{H}_\infty(\mathbf{e}|\mathbf{y}=\mathbf{sA}+\mathbf{e}) = -\log\left(\mathsf{E}_\mathbf{y}\left[\max_{\mathbf{e}^*}\Pr_{\mathbf{s},\mathbf{e}}[\mathbf{e}=\mathbf{e}^*|\mathbf{y}=\mathbf{sA}+\mathbf{e}]\right]\right),$$

where $\mathbf{e}^*$ is the point that maximizes the conditional probability $\Pr_{\mathbf{s},\mathbf{e}}[\mathbf{e}=\mathbf{e}^*|\mathbf{y}=\mathbf{sA}+\mathbf{e}]$. Given $\mathbf{A}$ and $\mathbf{y}$, when $\mathbf{s}^*\mathbf{A}$ is the closest lattice point to $\mathbf{y}$, $||\mathbf{e}|| = ||\mathbf{y}-\mathbf{s}^*\mathbf{A}||$ is minimized. As $\mathbf{e}$ is discrete Gaussian, it holds that $\mathbf{e}^* = \mathbf{y}-\mathbf{s}^*\mathbf{A}$. Events $\mathbf{e}=\mathbf{e}^*|\mathbf{y}=\mathbf{sA}+\mathbf{e}$ and $\mathbf{s}=\mathbf{s}^*|\mathbf{y}=\mathbf{sA}+\mathbf{e}$ are equivalent, as shown in Figure 1. Furthermore, it holds that $\Pr_{\mathbf{s},\mathbf{e}}[\mathbf{e}=\mathbf{e}^*|\mathbf{y}=\mathbf{sA}+\mathbf{e}] = \Pr[\mathbf{e}\in V]$ where V is the *Voronoïcell* of the lattice point $\mathbf{s}^*\mathbf{A}$. By Lemma 1 it holds that $\Pr(\mathbf{e}\in V) < 1/\rho_\sigma(\Lambda_q(\mathbf{A}))$. As $\mathbf{s}$ is drawn from the discrete Gaussian, we can quantify the conditional entropy of $\mathbf{s}$.

Lattice-based, more general leakage-resilient model

In Sect. "The leakage-resilient of LWE samples with discrete Gaussian secrets", we first quantify the leakage-resilient properties of LWE, where the secrets are drawn from a discrete Gaussian distribution. As the secrets are uniform in a small range, the situation is different. In Sect. "The leakage-resilient of LWE samples with bounded uniform secrets and Gaussian noise" we describe a lattice contained in $\frac{q}{d}\mathbb{Z}^m$, then prove the leakage-resilient property of the LWE samples on this lattice.

The leakage-resilient of LWE samples with discrete Gaussian secrets

When $\mathbf{s}$ is drawn from a discrete Gaussian distribution, we cannot directly apply the proof in Brakerski and

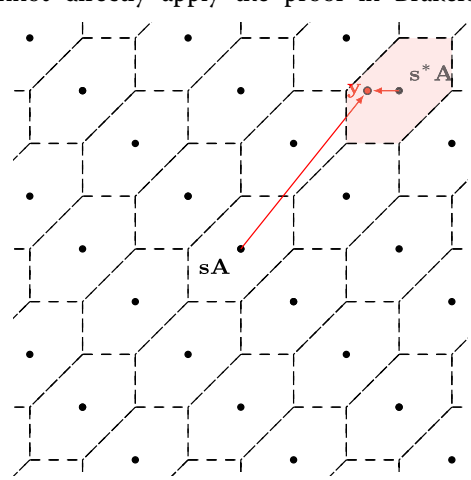

Fig. 1 $\mathbf{s}^*\mathbf{A}$ is the lattice point closest to $\mathbf{y}$, at this time $\mathbf{e}^*$ (the shorter red vector) happens to fall in the *Voronoïcell* (the red diamond area) of $\mathbf{s}^*\mathbf{A}$

Döttling (2018). At this time, we need to use the reduction technique Applebaum et al. (2009) from the LWE with discrete Gaussian secrets to the LWE with uniform secrets. Consider the following game.

- Alice picks matrix $\tilde{\mathbf{A}} = (\bar{\mathbf{A}}, \mathbf{A}') \in \mathbb{Z}_q^{n\times n} \times \mathbb{Z}_q^{n\times m}$, where $\bar{\mathbf{A}}$ is invertible, sends $\tilde{\mathbf{A}}$ it to Bob.
- After receiving $\tilde{\mathbf{A}}$, Bob generates $\mathbf{t} \leftarrow \mathbb{Z}_q^n$, $\mathbf{s} \leftarrow \mathcal{D}_{\mathbb{Z}^n,\sigma}$, $\mathbf{e} \leftarrow \mathcal{D}_{\mathbb{Z}^m,\sigma}$, $\bar{\mathbf{b}} = \mathbf{t}\bar{\mathbf{A}} - \mathbf{s}$, $\mathbf{b}' = \mathbf{t}\mathbf{A}' + \mathbf{e}$, $\mathbf{A} = -\bar{\mathbf{A}}^{-1}\mathbf{A}'$, sends $(\mathbf{A}, \mathbf{b} = \mathbf{b}' + \bar{\mathbf{b}}\mathbf{A})$ to Alice.

The above game essentially reduces discrete Gaussian LWE to standard LWE. Apparently $(\mathbf{A}, \mathbf{b} = \mathbf{s}\mathbf{A} + \mathbf{e})$ are the LWE samples with discrete Gaussian secrets, but $\mathbf{A}$ may not be uniform because $\tilde{\mathbf{A}}$ is chosen by Alice. Now, we quantify $\tilde{H}_\infty(\mathbf{s}|\mathbf{s}\mathbf{A} + \mathbf{e})$.

Theorem 1 *Let n, q, $m = O(n \log q)$ be integers, and $0 < \sigma < q/(2\sqrt{m+n})$. For the given matrix $\tilde{\mathbf{A}} = (\bar{\mathbf{A}}, \mathbf{A}') \in \mathbb{Z}_q^{n\times n} \times \mathbb{Z}_q^{n\times m}$ where $\bar{\mathbf{A}}$ is invertible, let $\mathbf{A} = -\bar{\mathbf{A}}^{-1}\mathbf{A}'$, $\mathbf{s} \leftarrow \mathcal{D}_{\mathbb{Z}^n,\sigma}$, $\mathbf{e} \leftarrow \mathcal{D}_{\mathbb{Z}^m,\sigma}$. It holds that*

$$\tilde{H}_\infty(\mathbf{s}|\mathbf{s}\mathbf{A} + \mathbf{e}) \geq -\log\left(\frac{1}{\rho_\sigma(\Lambda_q(\tilde{\mathbf{A}}))} + 2^{-(m+n)}\right).$$

Proof
Let $\tilde{\mathbf{e}} = (-\mathbf{s}, \mathbf{e})$, $\mathbf{t} \leftarrow \mathbb{Z}_q^n$, $\bar{\mathbf{b}} = \mathbf{t}\bar{\mathbf{A}} - \mathbf{s}$, $\mathbf{b}' = \mathbf{t}\mathbf{A}' + \mathbf{e}$, $\tilde{\mathbf{b}} = (\bar{\mathbf{b}}, \mathbf{b}')$. *According to the definition of average min-entropy, we have*

$$\begin{aligned}\tilde{H}_\infty(\tilde{\mathbf{e}}|\tilde{\mathbf{b}}) &= \tilde{H}_\infty(\tilde{\mathbf{e}}|\tilde{\mathbf{b}} = \mathbf{t}\tilde{\mathbf{A}} + \tilde{\mathbf{e}}) \\ &= -\log\left(\mathsf{E}_{\tilde{\mathbf{b}}}\left[\max_{\tilde{\mathbf{e}}^*} \Pr_{\mathbf{t},\tilde{\mathbf{e}}}[\tilde{\mathbf{e}} = \tilde{\mathbf{e}}^*|\tilde{\mathbf{b}} = \mathbf{t}\tilde{\mathbf{A}} + \tilde{\mathbf{e}}]\right]\right).\end{aligned}$$

Obviously, $\tilde{\mathbf{e}}$ that maximizes the conditional probability $\Pr[\tilde{\mathbf{e}} = \tilde{\mathbf{e}}^*|\tilde{\mathbf{b}} = \mathbf{t}\tilde{\mathbf{A}} + \tilde{\mathbf{e}}]$ must fall in the *Voronoï cell* of the lattice point that nearest to $\tilde{\mathbf{b}}$, that is $\tilde{\mathbf{e}}^* = \tilde{\mathbf{b}} - \mathbf{t}^*\tilde{\mathbf{A}}$ ($\mathbf{t}^*\tilde{\mathbf{A}}$ is the nearest lattice point to $\tilde{\mathbf{b}}$). By Theorem 3, it holds that $\Pr[\tilde{\mathbf{e}} = \tilde{\mathbf{e}}^*|\tilde{\mathbf{b}} = \mathbf{t}\tilde{\mathbf{A}} + \tilde{\mathbf{e}}] = \Pr[\tilde{\mathbf{e}} \mod q \in V]$, where $V \in \mathbb{Z}^{m+n}$ is the discretized *Voronoïcell* of $\Lambda_q(\tilde{\mathbf{A}})$. By Lemma 3, it holds that $||\tilde{\mathbf{e}}|| < \sigma \cdot \sqrt{m+n} < q/2$ except with probability $2^{-(m+n)}$, thus $\Pr[\tilde{\mathbf{e}} \mod q \in V] \leq \Pr[\tilde{\mathbf{e}} \in V] + 2^{-(m+n)}$. By Lemma 1, it holds that $\Pr[\tilde{\mathbf{e}} \in V] \leq \rho_\sigma(V)/\rho_\sigma(\mathbb{Z}^{m+n}) \leq 1/\rho_\sigma(\Lambda_q(\tilde{\mathbf{A}}))$, therefore $\Pr[\tilde{\mathbf{e}} \mod q \in V] \leq 1/\rho_\sigma(\Lambda_q(\tilde{\mathbf{A}})) + 2^{-(m+n)}$. We have

$$\begin{aligned}\tilde{H}_\infty(\tilde{\mathbf{e}}|\tilde{\mathbf{b}}) &= -\log\left(\mathsf{E}_{\tilde{\mathbf{b}}}[\Pr[\tilde{\mathbf{e}} \mod q \in V]]\right) \\ &= -\log\left(\Pr[\tilde{\mathbf{e}} \mod q \in V]\right) \\ &\geq -\log\left(\frac{1}{\rho_\sigma(\Lambda_q(\tilde{\mathbf{A}}))} + 2^{-(m+n)}\right).\end{aligned} \tag{1}$$

According to the chain rule of entropy: $H(X, Y) = H(X) + H(Y|X)$, we have

$$\tilde{H}_\infty((-\mathbf{s}, \mathbf{e})|\mathbf{s}\mathbf{A} + \mathbf{e}) = \tilde{H}_\infty(-\mathbf{s}, \mathbf{e}, \mathbf{s}\mathbf{A} + \mathbf{e}) - \tilde{H}_\infty(\mathbf{s}\mathbf{A} + \mathbf{e}). \tag{2}$$

As $\mathbf{A}$ is public $\tilde{H}_\infty(\mathbf{e}|(-\mathbf{s}, \mathbf{s}\mathbf{A} + \mathbf{e})) = 0$. Again by the chain rule we have

$$\tilde{H}_\infty(-\mathbf{s}, \mathbf{e}, \mathbf{s}\mathbf{A} + \mathbf{e}) = \tilde{H}_\infty(-\mathbf{s}, \mathbf{s}\mathbf{A} + \mathbf{e}). \tag{3}$$

Combining (2), (3) we have

$$\begin{aligned}\tilde{H}_\infty((-\mathbf{s}, \mathbf{e})|\mathbf{s}\mathbf{A} + \mathbf{e}) &= \tilde{H}_\infty(-\mathbf{s}, \mathbf{s}\mathbf{A} + \mathbf{e}) \\ &- \tilde{H}_\infty(\mathbf{s}\mathbf{A} + \mathbf{e}) = \tilde{H}_\infty(-\mathbf{s}|\mathbf{s}\mathbf{A} + \mathbf{e}).\end{aligned} \tag{4}$$

As $\mathbf{s}\mathbf{A} + \mathbf{e} = \mathbf{b}' + \bar{\mathbf{b}}\mathbf{A}$ we have

$$\tilde{H}_\infty((-\mathbf{s}, \mathbf{e})|\mathbf{s}\mathbf{A} + \mathbf{e}) \geq \tilde{H}_\infty((-\mathbf{s}, \mathbf{e})|\tilde{\mathbf{b}}). \tag{5}$$

Combining (1), (4), (5) we have

$$\tilde{H}_\infty(\mathbf{s}|\mathbf{s}\mathbf{A} + \mathbf{e}) \geq -\log\left(\frac{1}{\rho_\sigma(\Lambda_q(\tilde{\mathbf{A}}))} + 2^{-(m+n)}\right).$$

□

The leakage-resilient of LWE samples with bounded uniform secrets and Gaussian noise

Before presenting our results, we need to introduce the lattice on $\frac{q}{d}\mathbb{Z}^m$. Let $d, q \in \mathbb{Z}$ and $d \leq q$, $\mathbf{A} \in \frac{q}{d}\mathbb{Z}^{n\times m}$, $\mathbf{s} \in \mathbb{Z}_d^n$. Let

$$\Lambda_q(\mathbf{A}) = \{\mathbf{x} \in \frac{q}{d}\mathbb{Z}^m : \mathbf{x} = \mathbf{s}\mathbf{A} \mod q, \mathbf{s} \leftarrow \mathbb{Z}_d^n\}.$$

It is easy to verify that $\Lambda_q(\mathbf{A})$ forms a lattice, for any $\mathbf{x}_1$, $\mathbf{x}_2 \in \Lambda_q(\mathbf{A})$, let $\mathbf{x}_1 = \mathbf{s}_1\mathbf{A} \mod q$, $\mathbf{x}_2 = \mathbf{s}_2\mathbf{A} \mod q$, there exist $\mathbf{x}_3 \in \Lambda_q(\mathbf{A})$ satisfying $\mathbf{x}_3 = \mathbf{x}_1 + \mathbf{x}_2 \mod q$, where $\mathbf{s}_3 = \mathbf{s}_1 + \mathbf{s}_2 \mod d$, $\mathbf{x}_3 = \mathbf{s}_3\mathbf{A} \mod q$. That is, $\Lambda_q(\mathbf{A})$ is closed under addition modulo q, and is a discrete additive subgroup of $\frac{q}{d}\mathbb{Z}^m$.

It may be seen at a glance that $\Lambda_q(\mathbf{A})$ is isomorphic to the d-ary lattice (obtained by stretching d-ary lattice by a factor $\frac{q}{d}$). Such as for any $\mathbf{A} \in \frac{q}{d}\mathbb{Z}^{n\times m}$, let $\mathbf{A} = \frac{q}{d}\mathbf{A}'$, where $\mathbf{A}' \in \mathbb{Z}^{n\times m}$, there is a bijection ϕ between $\Lambda_d(\mathbf{A}') = \{\mathbf{x}' \in \mathbb{Z}^m : \mathbf{x}' = \mathbf{s}\mathbf{A}' \mod d, \mathbf{s} \leftarrow \mathbb{Z}_d^n\}$ and $\Lambda_q(\mathbf{A})$. For any $\mathbf{x}' \in \Lambda_d(\mathbf{A}')$, let $\mathbf{x}' = \mathbf{v} + d \cdot \mathbf{c}$, where $\mathbf{v} \in \mathbb{Z}_d^m$, $\mathbf{c} \in \mathbb{Z}^m$, its image in $\Lambda_q(\mathbf{A})$ is $\mathbf{x} = \frac{q}{d}\mathbf{v} + q \cdot \mathbf{c}$.

$$\phi: \quad \Lambda_d(\mathbf{A}') \to \Lambda_q(\mathbf{A})$$
$$\mathbf{v} + d \cdot \mathbf{c} \mapsto \frac{q}{d} \cdot \mathbf{v} + q \cdot \mathbf{c}.$$

For any $d \le q$, we provide the corresponding result when $\mathbf{s}$ is uniformly distributed on $\mathbb{Z}_d^n$ and $\mathbf{e}$ is sampled from discrete Gaussian distribution. As a compromise, the lattice $\Lambda_q(\mathbf{A})$ should also be adjusted accordingly.

Theorem 2 *Let n, d, q, $m = O(n \log d)$ be integers and $0 < d \le q, 0 < \sigma < d/\sqrt{m}$. For the given $\mathbf{A} \in \frac{q}{d}\mathbb{Z}_d^{n\times m}$, let $\mathbf{s} \leftarrow \mathbb{Z}_d^n$ and $\mathbf{e} \leftarrow \mathcal{D}_{\frac{q}{d}\mathbb{Z}^m,\sigma}$, then it holds that*

$$\tilde{H}_\infty(\mathbf{s}|\mathbf{sA} + \mathbf{e}) \ge -\log\left(\frac{1}{\rho_\sigma(\Lambda_q(\mathbf{A}))} + 2^{-m}\right).$$

If $d = q$, the above theorem degenerates into Lemma 3.2 in Brakerski and Döttling (2018). Its proof is the same as Brakerski and Döttling (2018); for the sake of completeness, we list it in Appendix B.

The hardness of entropy LWE

Before constructing our scheme, we first prove the hardness of distinguishing LWE samples with compromised keys from uniform distributions. That is, for LWE samples $(\mathbf{A}, \mathbf{b} = \mathbf{sA} + \mathbf{e})$, as long as the conditional entropy of $\mathbf{s}$ is large enough, then the hardness of distinguishing $(\mathbf{A}, \mathbf{b} = \mathbf{sA} + \mathbf{e})$ from uniform distributions is no less than distinguishing low-dimensional LWE samples from uniform distributions. This result sounds similar to the argument of entropy LWE at first glance. Brakerski and Döttling made a series of characterizations of entropy LWE in Brakerski and Döttling (2020a, 2020b), but their result cannot be applied to our scheme. This is because the conditional distribution in their result is restricted to $\mathbf{s}|\mathbf{s} + \mathbf{e}$. The conditional distribution of our Theorem 1 and Theorem 2 is restricted to $\mathbf{s}|\mathbf{sA} + \mathbf{e}$. Theorem 4 is more suitable for us because it does not restrict the conditional distribution of $\mathbf{s}$.

Goldwasser et al. (2010) proved the leakage-resilient property of such "weak" LWE samples $(\mathbf{A}, \mathbf{b} = \mathbf{sA} + \mathbf{e})$ where $\mathbf{s}$ is taken from $\{0,1\}^n$ (as the entropy of $\mathbf{s}$ is sufficient, it is no simpler than the low-dimensional standard LWE problem). It essentially utilizes the leakage-resilient property of the leftover hash lemma and reduces it to low-dimensional DLWE samples. Here, we prove a similar result based on the result of Theorem 4 where $\mathbf{s}$ is drawn from a discrete Gaussian distribution with bounded conditional entropy. This result will be used in the security proof of our scheme.

Lemma 4

Let n, m, q, r, λ be integers and q prime, $\mathcal{D}_{\mathbb{Z}^n,\sigma_1}$, $\mathcal{D}_{\mathbb{Z}^m,\sigma_2}$ be the discrete Gaussian distribution over $\mathbb{Z}^n$ and $\mathbb{Z}^m$ with standard deviation σ_1, σ_2 respectively satisfying $(n\sigma_1^2)/\sigma_2 = \mathsf{negl}(\lambda)$. Let $\mathbf{A} \leftarrow \mathbb{Z}_q^{n\times m}$, $\mathbf{s} \leftarrow \mathcal{D}_{\mathbb{Z}^n,\sigma_1}$, $\mathbf{e} \leftarrow \mathcal{D}_{\mathbb{Z}^m,\sigma_2}$ and $\mathsf{aux}(\mathbf{s})$ be the auxiliary information on $\mathbf{s}$. Let $\mathcal{D}_0 = (\mathbf{A}, \mathbf{sA} + \mathbf{e}, \mathsf{aux}(\mathbf{s}))$, $\mathcal{D}_1 = (U(\mathbb{Z}_q^{n\times m} \times \mathbb{Z}_q^m), \mathsf{aux}(\mathbf{s}))$. Let $r = (n - 2\lambda)/\log q$, assuming $\tilde{H}_\infty(\mathbf{s}|\mathsf{aux}(\mathbf{s})) \ge n$, if there is an Algorithm can distinguish $\mathcal{D}_0$ from $\mathcal{D}_1$, then it can distinguish the standard LWE samples with dimensional r from uniform distribution.

Proof

By the $\mathsf{DLWE}_{r,n,q,\sigma_1}$ assumption, we can replace $\mathbf{A}$ with $\mathbf{BC} + \mathbf{E}$, where $\mathbf{B} \leftarrow \mathbb{Z}_q^{n\times r}, \mathbf{C} \leftarrow \mathbb{Z}_q^{r\times m}, \mathbf{E} \leftarrow \mathcal{D}_{\mathbb{Z}^{n\times m},\sigma_1}$, thus

$$\mathbf{sA} + \mathbf{e} = \mathbf{sBC} + \mathbf{sE} + \mathbf{e} \approx_s \mathbf{sBC} + \mathbf{e},$$

as $(n\sigma_1^2)/\sigma_2 = \mathsf{negl}(\lambda)$. Let $\epsilon = 2^{-\lambda}$, $\delta = n \log \sigma_1 - n$, by the assumption $\tilde{H}_\infty(\mathbf{s}|\mathsf{aux}(\mathbf{s})) \ge n$ and Theorem 4 we have

$$(\mathbf{B}, \mathbf{sB}, \mathsf{aux}(\mathbf{s})) \approx_s (U(\mathbb{Z}_q^{n\times r} \times \mathbb{Z}_q^r), \mathsf{aux}(\mathbf{s})).$$

Let $\mathbf{t} = \mathbf{sB}$ we have $\mathbf{sA} + \mathbf{e} \approx_s \mathbf{tC} + \mathbf{e}$ which are the LWE samples with dimensional r and standard deviation $\sigma_2 = (n\sigma_1^2)/\mathsf{negl}(\lambda)$. If there is an Algorithm can distinguish $\mathcal{D}_0$ from $\mathcal{D}_1$, then it can distinguish the LWE samples with dimensional r from uniform distribution. □

Optimized multi-key fully homomorphic encryption scheme

In 2012, López-Alt et al. (2012) proposed the concept of Multi-key fully homomorphic encryption, and constructed the first MKFHE based on the NTRU encryption scheme. It was an extension of the single-key fully homomorphic scheme (supports homomorphic operations between ciphertexts encrypted with different public keys). After that, by introducing CRS, Clear and McGoldrick (2015), Mukherjee and Wichs (2016), Peikert and Shiehian (2016) constructed the GSW type MKFHE. Chen et al. (2017) and Chen et al. (2019) constructed the MKFHE based on RLWE and applied it to privacy-preserving neural network training with multiple parties.

The work Brakerski et al. (2017) was the first MKFHE scheme that does not introduce CRS. By the leftover hash lemma, it proved the security of its scheme as the entropy of the private key is sufficient (the tradeoff is that the length of the private key increases with the amount of leakage). Ananth et al. (2020) removed CRS from a higher dimension; instead of using the leftover hash lemma or regularity lemma, they based on *Multiparty homomorphic encryption* and modified the initialization method of its root node to achieve this goal.

In this section, we first describe our MKFHE and then give the security proof. All GSW-based MKFHE require a *ciphertext expansion* operation on the initial ciphertext to support homomorphic evaluation. The *ciphertext expansion* operation of our scheme is similar to that of the previous scheme, so we put this part at the end.

*Scheme#*1: a "GSW-style" MKFHE in the plain model

As an application of Theorem 1 and Theorem 4, we give an optimized MKFHE scheme based on Brakerski et al. (2017). It must be pointed out that such optimization can also be applied to Clear and McGoldrick (2015); Mukherjee and Wichs (2016); Peikert and Shiehian (2016); Dai et al. (2022) and other GSW-based MKFHE (constructed on $\mathbb{Z}$, can use the leftover hash lemma to remove CRS). We choose Brakerski et al. (2017) as an example because it requires fewer changes, and the improved result is better. The improved scheme is defined as follows

- $\mathsf{pp} \leftarrow \mathsf{setup}(1^\lambda, 1^k, 1^L)$: On input security parameter λ, users number $k = \mathsf{poly}(\lambda)$, circuit depth L, let $n = \mathsf{poly}(\lambda)$, $q = 2^{O(L)} \cdot \mathsf{poly}(\lambda)$, $l = \lceil \log q \rceil$, $m = (n+1)l$ be an integers, and q prime. Let σ_1, $\sigma_2 \in \mathbb{R}^+$ and satisfying $(m\sigma_1)/\sigma_2 = \mathsf{negl}(\lambda)$. Suitable choosing the above parameters to make the $\mathsf{DLWE}_{n,m,q,\sigma_1}$ problem is infeasible, output $\mathsf{pp} = (k, n, m, q, \sigma_1, \sigma_2)$.
- $(\mathsf{pk}_i, \mathsf{sk}_i) \leftarrow \mathsf{Gen}(\mathsf{pp}, i)$: Input pp and the user ID i, output the key pair $(\mathsf{pk}_i, \mathsf{sk}_i)$ of party i, where $\mathsf{pk}_i = (\mathbf{A}_i, \mathbf{b}_{i,i})$, $\mathbf{A}_i \leftarrow \mathbb{Z}_q^{n\times m}$, $\mathbf{s}_i \leftarrow \mathcal{D}_{\mathbb{Z}^n,\sigma_1}$, $\mathbf{e}_i \leftarrow \mathcal{D}_{\mathbb{Z}^m,\sigma_1}$, $\mathbf{b}_{i,i} = \mathbf{s}_i\mathbf{A}_i + \mathbf{e}_i$, $\mathsf{sk}_i = (\mathbf{s}_i, -1)$.
- $\mathsf{Auxk}_i \leftarrow \mathsf{AuxKeyGen}(\mathsf{sk_i}, \{\mathsf{pk_j}\}_{\mathsf{j\in[k]/i}})$: Input the private key sk_i of party i and other parties public keys $\{\mathsf{pk}_j\}_{j\in[k]/i}$, output the Auxiliary key (for *ciphertext expansion*) $\mathsf{Auxk}_i = \{\mathbf{b}_{i,j}\}_{j\in[k]/i}$ of party i where $\mathbf{b}_{i,j} = \mathbf{s}_i\mathbf{A}_j + \mathbf{e}_j$, $\mathbf{e}_j \leftarrow \mathcal{D}_{\mathbb{Z}^m,\sigma_1}$.
- $\mathbf{C}_i \leftarrow \mathsf{Enc}(\mathsf{pk}_i, u)$: Input public key pk_i, a plaintext $u \in \{0,1\}$, output ciphertext

$$\mathbf{C}_i = \begin{pmatrix} \mathbf{A}_i \\ \mathbf{b}_{i,i} \end{pmatrix} \mathbf{R} + \begin{pmatrix} \mathbf{0} \\ \mathbf{e}' \end{pmatrix} + u\mathbf{G},$$

  where $\mathbf{e}' \leftarrow \mathcal{D}_{\mathbb{Z}^m,\sigma_2}$, $\mathbf{R} \leftarrow \{0,1\}^{m\times m}$, $\mathbf{G}$ is a gadget matrix as defined in Preliminary.
- $u \leftarrow \mathsf{Dec}(\mathsf{sk}, \mathbf{C})$: Input ciphertext $\mathbf{C}$, private key sk, let $\mathbf{t} = \mathsf{sk}$, $\mathbf{w}^T = (0, \cdots, 0, \lfloor \frac{q}{2} \rceil) \in \mathbb{Z}_q^{n+1}$, $\gamma = \mathbf{t} \cdot \mathbf{C}\mathbf{G}^{-1}(\mathbf{w}^T)$, output $u = \lfloor \frac{\gamma}{q/2} \rceil$.

Security under semi-malicious adversary

We note that the auxiliary key of party i is $\mathsf{Auxk}_i = \{\mathbf{b}_{i,j} = \mathbf{s}_i\mathbf{A}_j + \mathbf{e}_j\}_{j\in[k]/i}$ where $\{\mathbf{A}_j\}_{j\in[k]/i}$ are generated by the other $k-1$ parties. Under the semi-honest adversary, $\{\mathbf{A}_j\}_{j\in[k]/i}$ are uniformly distributed over $\mathbb{Z}_q^{n\times m}$. Under the $\mathsf{DLWE}_{n,m,q,\sigma_1}$ assumption, Auxk_i is indistinguishable from the uniform distribution, and the security of the scheme is now obvious.

However, under the semi-malicious adversary, $\{\mathbf{A}_j\}_{j\in[k]/i}$ may not be uniform, and the conditional distributions $\mathbf{s}_i|\mathsf{Auxk}_i$ and $\mathbf{s}_i$ may differ significantly. Next, we prove that as long as the conditional entropy $\tilde{H}_\infty(\mathbf{s}_i|\mathsf{Auxk}_i)$ is large enough, the scheme is still secure.

Claim 1

Let $(\mathbf{A}_i, \mathbf{b}_{i,i})$, Auxk_i, $\mathbf{C}_i$ *be the public key, auxiliary key and ciphertext of party* i *in Scheme#*1. *Let* $\mathcal{D}_0 = ((\mathbf{A}_i, \mathbf{b}_{i,i}), \mathsf{Auxk}_i, \mathbf{C}_i)$ *and* $\mathcal{D}_1 = ((\mathbf{A}_i, \mathbf{b}_{i,i}), \mathsf{Auxk}_i, \mathcal{U}(\mathbb{Z}_q^{(n+1)\times m}))$. *Let* $r = (n-2\lambda)/\log q$, *assuming* $\tilde{H}_\infty(\mathbf{s}_i|\mathsf{Auxk}_i) \geq n$, *If there is an adversary can distinguish between* $\mathcal{D}_0$ *and* $\mathcal{D}_1$, *then he also distinguish the LWE samples with dimensional* r *from uniform distribution.*

Proof

Let $\mathbf{C}_i = \begin{pmatrix} \mathbf{C}_0 \\ \mathbf{c}_1 \end{pmatrix} = \begin{pmatrix} \mathbf{A}_i \\ \mathbf{b}_{i,i} \end{pmatrix} \mathbf{R} + \begin{pmatrix} \mathbf{0} \\ \mathbf{e}' \end{pmatrix}$ *as* $\mathbf{b}_{i,i} = \mathbf{s}_i\mathbf{A}_i + \mathbf{e}_i$ *it holds that* $\mathbf{c}_1 = \mathbf{s}_i\mathbf{A}_i\mathbf{R} + \mathbf{e}_i\mathbf{R} + \mathbf{e}' = \mathbf{s}_i\mathbf{C}_0 + \mathbf{e}_i\mathbf{R} + \mathbf{e}'$. *By our parameter settings we have* $||\mathbf{e}_i\mathbf{R}/\mathbf{e}'|| = \mathsf{negl}(\lambda)$ *thus*

$$\begin{aligned} \mathcal{D}_0 &= \left(\mathbf{A}_i,\ \mathbf{b}_{i,i},\ \mathsf{Auxk}_i, \begin{pmatrix} \mathbf{C}_0 \\ \mathbf{c}_1 \end{pmatrix}\right) \\ &\approx_s \left(\mathbf{A}_i,\ \mathbf{b}_{i,i},\ \mathsf{Auxk}_i, \begin{pmatrix} \mathbf{C}_0 \\ \mathbf{s}_i\mathbf{C}_0 + \mathbf{e}' \end{pmatrix}\right). \end{aligned}$$

Using the leftover hash lemma with $\mathbf{A}_i$ as seed and $\mathbf{R}$ as source we have $(\mathbf{A}_i, \mathbf{C}_0) \approx_s (\mathbf{A}_i, \mathbf{Z})$ where $\mathbf{Z} \leftarrow \mathbb{Z}_q^{n\times m}$ thus

$$\begin{aligned} &\left(\mathbf{A}_i, \mathbf{b}_{i,i},\ \mathsf{Auxk}_i, \begin{pmatrix} \mathbf{C}_0 \\ \mathbf{s}_i\mathbf{C}_0 + \mathbf{e}' \end{pmatrix}\right) \\ &\approx_s \left(\mathbf{A}_i, \mathbf{b}_{i,i},\ \mathsf{Auxk}_i, \begin{pmatrix} \mathbf{Z} \\ \mathbf{s}_i\mathbf{Z} + \mathbf{e}' \end{pmatrix}\right). \end{aligned}$$

We note that $\mathbf{Z}$ is independent of $\mathbf{s}_i$ as $\mathbf{C}_0$ is generated after $\mathbf{s}_i$ and $\mathbf{b}_{i,j}$. By the assumption $\tilde{H}_\infty(\mathbf{s}_i|\mathsf{Auxk}_i) \geq n$ and Lemma 4 we have

$$\left(\mathbf{A}_i, \mathbf{b}_{i,i}, \mathsf{Auxk}_i, \begin{pmatrix}\mathbf{Z}\\ \mathbf{s}_i\mathbf{Z}+\mathbf{e}'\end{pmatrix}\right) \approx_c \left(\mathbf{A}_i, \mathbf{b}_{i,i}, \mathsf{Auxk}_i, \begin{pmatrix}\mathbf{Z}\\ \mathbf{z}\end{pmatrix}\right) = \mathcal{D}_1,$$

where $\mathbf{z} \leftarrow \mathbb{Z}_q^m$. □

Note that the premise of the above result is that $\tilde{H}_\infty(\mathbf{s}_i|\{\mathbf{b}_{i,j}\}_{j\in[k]/i}) \geq n$ where $\mathbf{b}_{i,j} = \mathbf{s}_i\mathbf{A}_j + \mathbf{e}_j$. Without loss of generality, let $i = 1$ we have

$$(\mathbf{b}_{1,2}, \mathbf{b}_{1,3}, \cdots, \mathbf{b}_{1,k}) = \mathbf{s}_1(\mathbf{A}_2, \mathbf{A}_3, \cdots, \mathbf{A}_k) + (\mathbf{e}_2, \mathbf{e}_3, \cdots, \mathbf{e}_k).$$

Let $\mathbf{A} = (\mathbf{A}_2, \mathbf{A}_3, \cdots, \mathbf{A}_k)$, $\tilde{\mathbf{A}} = (\mathbf{I}_n, \mathbf{A})$, where $\mathbf{I}_n$ is the dimensional n identity matrix. Let $\mathbf{e} = (\mathbf{e}_2, \mathbf{e}_3, \cdots, \mathbf{e}_k)$, by Theorem 1 if $0 < \sigma_1 < q/(2\sqrt{m(k-1)+n})$ we have

$$\tilde{H}_\infty(\mathbf{s}_i|\mathbf{s}_i\mathbf{A}+\mathbf{e}) \geq -\log\left(\frac{1}{\rho_{\sigma_1}(\Lambda_q(\tilde{\mathbf{A}}))} + 2^{-m(k-1)-n}\right). \tag{6}$$

Let $\gamma > 0$ be real $\mathcal{B}$ be a sphere of radius 1, by Lemma 2 if $\mathsf{rank}(\Lambda_q(\tilde{\mathbf{A}}) \bigcap \gamma\mathcal{B}) \geq n/2$ and $\sigma_1 > 5\gamma$, it holds that $\rho_{\sigma_1}(\Lambda_q(\tilde{\mathbf{A}})) > 5^{n/2}$ satisfying $1/\rho_{\sigma_1}(\Lambda_q(\tilde{\mathbf{A}})) \leq 2^{-n} - 2^{-m(k-1)-n}$, thus $\tilde{H}_\infty(\mathbf{s}_i|\mathbf{s}_i\mathbf{A}+\mathbf{e}) \geq n$. We observe from Ananth et al. (2021) that one way to make $\mathsf{rank}(\Lambda_q(\tilde{\mathbf{A}}) \bigcap \gamma\mathcal{B}) \geq n/2$ is to have $\tilde{\mathbf{A}}$ structured as

$$\begin{aligned}\tilde{\mathbf{A}} &= (\mathbf{I}_n, \mathbf{A}) = (\mathbf{I}_n, \mathbf{A}_2, \mathbf{A}_3, \cdots, \mathbf{A}_k)\\ &= \begin{pmatrix}\mathbf{I}_n, & \mathbf{B}_2 & \mathbf{B}_3 & \cdots & \mathbf{B}_k\\ & \mathbf{S}\mathbf{B}_2+\mathbf{E}_2, & \mathbf{S}\mathbf{B}_3+\mathbf{E}_3, & \cdots, & \mathbf{S}\mathbf{B}_k+\mathbf{E}_k\end{pmatrix},\end{aligned}$$

where $\mathbf{B}_i \leftarrow \mathbb{Z}_q^{\frac{n}{2}\times m}$, $\mathbf{S} \leftarrow \mathcal{D}_{\mathbb{Z}^{\frac{n}{2}\times\frac{n}{2}},\sigma_3}$, $\mathbf{E}_i \leftarrow \mathcal{D}_{\mathbb{Z}^{\frac{n}{2}\times m},\sigma_3}$ and $\sqrt{2n} < \sigma_3 < \gamma/\sqrt{m(k-1)+n}$. Thus it holds that

$$\begin{pmatrix}\mathbf{I}_{\frac{n}{2}} & \\ -\mathbf{S}, & \mathbf{I}_{\frac{n}{2}}\end{pmatrix} \cdot \tilde{\mathbf{A}} = \begin{pmatrix}\mathbf{I}_{\frac{n}{2}} & \mathbf{B}_2, & \mathbf{B}_3, & \cdots, & \mathbf{B}_k\\ -\mathbf{S}, & \mathbf{I}_{\frac{n}{2}}, & \mathbf{E}_2, & \mathbf{E}_3, & \cdots, & \mathbf{E}_k\end{pmatrix} \in \Lambda_q(\tilde{\mathbf{A}}),$$

where $\mathbf{I}_{\frac{n}{2}}$ is the dimensional $n/2$ identity matrix. Let $\mathbf{V} = (-\mathbf{S}, \mathbf{I}_{\frac{n}{2}}, \mathbf{E}_2, \mathbf{E}_3, \cdots, \mathbf{E}_k)$ be the lower half of the above matrix, it holds that $\mathbf{V} \in \Lambda_q(\tilde{\mathbf{A}})$. By Lemma 3, the length of all row vectors of $\mathbf{V}$ does not exceed γ, and $\mathsf{rank}(\mathbf{V}) = n/2$. Thus $\mathsf{rank}(\Lambda_q(\tilde{\mathbf{A}}) \bigcap \gamma\mathcal{B}) \geq n/2$.

Put things together

We bring together the previous parameter requirements, in particular the range of standard deviations for several discrete Gaussian distributions. For (6) holds, we need $0 < \sigma_1 < q/(2\sqrt{m(k-1)+n})$. To make $\mathsf{rank}(\Lambda_q(\tilde{\mathbf{A}}) \bigcap \gamma\mathcal{B}) \geq n/2$ and $\mathbf{A}$ looks random, we need $\sigma_1 > 5\gamma$ and $\sqrt{2n} < \sigma_3 \leq \gamma/\sqrt{m(k-1)+n}$. In Lemma 4, we need $(n\sigma_1^2)/\sigma_2 = \mathsf{negl}(\lambda)$. To sum up, we get the standard deviation $\sigma_1, \sigma_2, \sigma_3$ respectively as follows

$$5\sqrt{2mn(k-1)+2n^2} < \sigma_1 < q/(2\sqrt{m(k-1)+n}),$$
$$\sigma_2 = (n\sigma_1^2)/\mathsf{negl}(\lambda), \quad \sigma_3 \geq \sqrt{2n}.$$

Ciphertext expansion

In order to convert the ciphertext under different keys into ciphertext under the same key, MKFHE with GSW-type ciphertext needs to undergo a *ciphertext expansion* operation. A typical *ciphertext expansion* is the masking scheme defined in Mukherjee and Wichs (2016); Peikert and Shiehian (2016); Clear and McGoldrick (2015), for any ciphertext $\mathbf{C} = \begin{pmatrix}\mathbf{A}\\ \mathbf{b}\end{pmatrix}\mathbf{R} + u\mathbf{G}$ (the private key is $\mathbf{t}$) to be expanded, input any $\mathbf{v} \in \mathbb{Z}_q^m$ and the ciphertext of the random matrix $\mathbf{R}$, the masking scheme applies the homomorphic property of the GSW scheme to output the ciphertext $\mathbf{X}$ of $\mathbf{vR}$ under $\mathbf{t}$ (satisfying $\mathbf{tX} \approx \mathbf{vR}$), which is used to eliminate the redundant items produced by decrypting $\mathbf{C}$ with the other party's private key.

We note that the above masking scheme works for our variant as well, simply because the encryption and decryption formulas are identical and all follow the GSW ciphertext structure. Below, for the sake of completeness, we informally describe the process of *ciphertext expansion*.

A masking scheme for GSW ciphertext

There exist a pair of algorithm (UniEnc, Extend), defined in Mukherjee and Wichs (2016); Clear and McGoldrick (2015) adapted to our scheme:

- $\mathsf{UniEnc}(\mathbf{R}, \mathbf{t})$: On input the matrix $\mathbf{R}$ in $\mathbf{C}$ and the public key $\mathbf{t}$, it output $\mathcal{U}$ the ciphertext of $\mathbf{R}$.
- $\mathsf{Extend}(\mathcal{U}, \mathbf{v})$: On input $\mathcal{U}$ and any $\mathbf{v} \in \mathbb{Z}_q^m$, it output $\mathbf{X} \in \mathbb{Z}_q^{(n+1)\times m}$.

It holds that $\mathbf{X}$ is the ciphertext of $\mathbf{vR}$ satisfying $\mathbf{tX} \approx \mathbf{vR}$. Without loss of generality, let $k = 2$, $\bar{\mathbf{C}} = \begin{pmatrix}\mathbf{C} & \mathbf{X}\\ & \mathbf{C}\end{pmatrix} \in \mathbb{Z}_q^{2(n+1)\times 2m}$ be the expanded ciphertext of our scheme under public key $\mathbf{t}_1$, let $\mathbf{t}_2$ be another public key, it holds that $(\mathbf{t}_1, \mathbf{t}_2)\bar{\mathbf{C}} = (\mathbf{t}_1\mathbf{C}, \mathbf{t}_1\mathbf{X} + \mathbf{t}_2\mathbf{C}) \approx (u\mathbf{t}_1\mathbf{G}, \mathbf{t}_1\mathbf{X} + (\mathbf{b}_{2,1} - \mathbf{b}_{1,1})\mathbf{R} + u\mathbf{t}_2\mathbf{G})$. To eliminate $(\mathbf{b}_{2,1} - \mathbf{b}_{1,1})\mathbf{R}$ we can set $\mathbf{v} = \mathbf{b}_{1,1} - \mathbf{b}_{2,1}$ where $\mathbf{b}_{2,1}$ is

the auxiliary key of party 2, then it holds that $\mathbf{t}_1\mathbf{X} \approx (\mathbf{b}_{1,1} - \mathbf{b}_{2,1})\mathbf{R}$, thus $(\mathbf{t}_1, \mathbf{t}_2)\bar{\mathbf{C}} \approx u(\mathbf{t}_1, \mathbf{t}_2)\begin{pmatrix}\mathbf{G} & \\ & \mathbf{G}\end{pmatrix}$. Thus $\bar{\mathbf{C}} = \begin{pmatrix}\mathbf{C} & \mathbf{X} \\ & \mathbf{C}\end{pmatrix}$ is the expanded ciphertext of $\mathbf{C}$ with only two parties. The above process can be extended to k parties. At this time, the expanded ciphertext of $\mathbf{C}$ is

$$\bar{\mathbf{C}} = \begin{pmatrix}\mathbf{C} & \mathbf{X}_1 & \cdots & \mathbf{X}_{k-1} \\ & \mathbf{C} & & \\ & & \cdots & \\ & & & \mathbf{C}\end{pmatrix} \in \mathbb{Z}_q^{k(n+1)\times km},$$

where the private key is $(\mathbf{t}_1, \cdots, \mathbf{t}_k)$ and $\{\mathbf{X}_i\}_{i\in[k-1]}$ satisfying $\mathbf{t}_1\mathbf{X}_i \approx (\mathbf{b}_{1,1} - \mathbf{b}_{i+1,1})\mathbf{R}$.

Homomorphic addition and multiplication

Let $\bar{\mathbf{C}}_1, \bar{\mathbf{C}}_2$ be the expanded ciphertext, $\bar{\mathbf{t}} = (\mathbf{t}_1, \mathbf{t}_2, \cdots, \mathbf{t}_k)$ and $\bar{\mathbf{G}} = \begin{pmatrix}\mathbf{G} & & & \\ & \mathbf{G} & & \\ & & \cdots & \\ & & & \mathbf{G}\end{pmatrix}$

- $\mathbf{C}_{\mathsf{add}} \leftarrow \mathsf{Add}(\bar{\mathbf{C}}_1, \bar{\mathbf{C}}_2)$: Input ciphertext $\bar{\mathbf{C}}_1$, $\bar{\mathbf{C}}_2$, output $\mathbf{C}_{\mathsf{add}} = \bar{\mathbf{C}}_1 + \bar{\mathbf{C}}_2$.
- $\mathbf{C}_{\mathsf{mult}} \leftarrow \mathsf{mult}(\bar{\mathbf{C}}_1, \bar{\mathbf{C}}_2)$: Input ciphertext $\bar{\mathbf{C}}_1$, $\bar{\mathbf{C}}_2$, output $\mathbf{C}_{\mathsf{mult}} = \bar{\mathbf{C}}_1 \cdot \bar{\mathbf{G}}^{-1}(\bar{\mathbf{C}}_2)$.

It holds that $\bar{\mathbf{t}} \cdot \mathbf{C}_{\mathsf{add}} \approx (u_1 + u_2)\bar{\mathbf{t}}\bar{\mathbf{G}}$ and $\bar{\mathbf{t}} \cdot \mathbf{C}_{\mathsf{mult}} \approx u_1u_2\bar{\mathbf{t}}\bar{\mathbf{G}}$.

Scheme#2: a "GSW-style" MKFHE in the plain model over $\frac{q}{d}\mathbb{Z}^m$

As an application of Theorem 2 we construct the MKFHE scheme over $\frac{q}{d}\mathbb{Z}^m$. Similar to *Scheme*#1, *Scheme*#2 also has a GSW-type ciphertext, but *Scheme*#1 is more natural and follows common practices. However, when the private key is taken from a uniform distribution on $\mathbb{Z}_d^n$ (for any $d < q$), in order to remove CRS using the result of Theorem 2, we have to construct the scheme over $\frac{q}{d}\mathbb{Z}^m$. *Scheme*#2 is defined as follows.

- $\mathsf{pp} \leftarrow \mathsf{setup}(1^\lambda, 1^k, 1^L)$: On input security parameter λ, users number $k = \mathsf{poly}(\lambda)$, circuit depth L, let $n = \mathsf{poly}(\lambda)$ be an integer, $d = 2^{O(\lambda L)}$ be a prime, $l = \lceil \log d \rceil$, $m = (n+1)l$, $q = d \cdot \mathsf{poly}(\lambda)$, $\sigma_1, \sigma_2 \in \mathbb{R}^+$ and $(m\sigma_1)/\sigma_2 = \mathsf{negl}(\lambda)$. Suitable choosing the above parameters to make $\mathsf{rantional\text{-}DLWE}_{n,m,d,q,\sigma_1}$ (defined in D.1) is infeasible, output $\mathsf{pp} = (k, n, m, d, q, \sigma_1, \sigma_2)$.
- $(\mathsf{pk}_i, \mathsf{sk}_i) \leftarrow \mathsf{Gen}(\mathsf{pp}, i)$: Input pp, user ID i, output the key pair $(\mathsf{pk}_i, \mathsf{sk}_i)$ of party i, where $\mathsf{pk}_i = (\mathbf{A}_i, \mathbf{b}_{i,i})$, $\mathbf{A}_i \leftarrow \frac{q}{d}\mathbb{Z}_d^{n\times m}$, $\mathbf{s}_i \leftarrow \mathbb{Z}_d^n$, $\mathbf{e} \leftarrow \mathcal{D}_{\frac{q}{d}\mathbb{Z}^m,\sigma_1}$, $\mathbf{b}_{i,i} = \mathbf{s}_i\mathbf{A}_i + \mathbf{e}_i \mod q$, $\mathsf{sk}_i = (\mathbf{s}_i, -1)$.
- $\mathsf{Auxk}_i \leftarrow \mathsf{Auxiliary\ KeyGen}(\mathsf{sk_i}, \{\mathsf{pk_j}\}_{\mathsf{j}\in[\mathsf{k}]/\mathsf{i}})$: Input the private key sk_i of party i and other parties public keys $\{\mathsf{pk}_j\}_{j\in[k]/i}$, output the Auxiliary key (for *ciphertext expansion*) $\mathsf{Auxk}_i = \{\mathbf{b}_{i,j}\}_{j\in[k]/i}$ of party i where $\mathbf{b}_{i,j} = \mathbf{s}_i\mathbf{A}_j + \mathbf{e}_j$.
- $\mathbf{C}_i \leftarrow \mathsf{Enc}(\mathsf{pk}_i, u_i)$: Input public key pk_i, a plaintext $u_i \in \{0, 1\}$, output ciphertext $\mathbf{C}_i = \begin{pmatrix}\mathbf{A}_i \\ \mathbf{b}_{i,i}\end{pmatrix} \cdot \mathbf{R} + \begin{pmatrix}\mathbf{0} \\ \mathbf{e}'\end{pmatrix} + u_i\mathbf{G}$ where $\mathbf{e}' \leftarrow \mathcal{D}_{\frac{q}{d}\mathbb{Z}^m,\sigma_2}$, $\mathbf{R} \leftarrow \{0, 1\}^{m\times m}$, $\mathbf{G}$ is a gadget matrix as defined in Preliminary.
- $u \leftarrow \mathsf{Dec}(\mathsf{sk}, \mathbf{C})$: Input ciphertext $\mathbf{C}$, private key sk, let $\mathbf{t} = \mathsf{sk}$, $\mathbf{w}^T = (0, \cdots, 0, \lfloor \frac{d}{2} \rceil \cdot \frac{q}{d}) \in \frac{q}{d}\mathbb{Z}_d^{n+1}$, $\gamma = \mathbf{t} \cdot \mathbf{C}\mathbf{G}^{-1}(\mathbf{w}^T)$, output $u = \lfloor \frac{\gamma}{q/2} \rceil$.

Security under semi-malicious adversary

Similar to *Scheme*#2, under the semi-honest adversary, $\{\mathbf{A}_j\}_{j\in[k]/i}$ are uniformly distributed over $\frac{q}{d}\mathbb{Z}_d^{n\times m}$, further by the $\mathsf{rational\text{-}DLWE}_{n,m,d,q,\sigma_1}$ assumption, Auxk_i is indistinguishable from the uniform distribution. Similarly, under a semi-malicious adversary, we need to assume $\tilde{H}_\infty(\mathbf{s}_i|\{\mathbf{b}_{i,j}\}_{j\in[k]/i})$ is large enough to prove security. We have the following result.

Claim 2

Let $(\mathbf{A}_i, \mathbf{b}_{i,i})$, $\mathbf{s}_i$, Auxk_i, $\mathbf{C}_i$ *be the public key, private key, auxiliary key and ciphertext of party i in Scheme*#2. *Let* $r = (n - \omega(\log n))/\log d$ *be integer,* $(\mathbf{B}, \mathbf{t})$ *be the public key of dual-Regev encryption where* $\mathbf{B} \leftarrow \frac{q}{d}\mathbb{Z}_d^{n\times r}$, $\mathbf{t} = \mathbf{s}_i\mathbf{B} \mod q$. *Let* $\mathcal{D}_0 = ((\mathbf{A}_i, \mathbf{b}_{i,i}), \mathsf{Auxk}_i, \mathbf{C}_i)$ *and* $\mathcal{D}_1 = ((\mathbf{A}_i, \mathbf{b}_{i,i}), \mathsf{Auxk}_i, \mathcal{U}(\mathbb{Z}_q^{(n+1)\times m}))$. *Assuming* $\tilde{H}_\infty(\mathbf{s}_i|\mathsf{Auxk}_i) \geq n$ *and dual-Regev encryption is circular security with public key* $(\mathbf{B}, \mathbf{t})$, *then it holds that* $\mathcal{D}_0$ *and* $\mathcal{D}_1$ *are (jointly) computational indistinguishable.*

Proof

Let $\mathbf{C} = \begin{pmatrix}\mathbf{C}_0 \\ \mathbf{c}_1\end{pmatrix} = \begin{pmatrix}\mathbf{A}_i \\ \mathbf{b}_{i,i}\end{pmatrix}\mathbf{R} + \begin{pmatrix}\mathbf{0} \\ \mathbf{e}'\end{pmatrix}$, *for* $\mathbf{b}_{i,i} = \mathbf{s}_i\mathbf{A}_i + \mathbf{e}_i$, *it holds that* $\mathbf{c}_1 = \mathbf{s}_i\mathbf{A}_i\mathbf{R} + \mathbf{e}_i\mathbf{R} + \mathbf{e}' = \mathbf{s}_i\mathbf{C}_0 + \mathbf{e}_i\mathbf{R} + \mathbf{e}'$. *By our parameter settings, we have* $||\mathbf{e}_i\mathbf{R}/\mathbf{e}'|| = \mathsf{negl}(\lambda)$, *thus*

$$\left(\mathbf{A}_i, \{\mathbf{b}_{i,j}\}_{j\in[k]/i}, \begin{pmatrix}\mathbf{C}_0 \\ \mathbf{c}_1\end{pmatrix}\right) \approx_s \left(\mathbf{A}_i, \{\mathbf{b}_{i,j}\}_{j\in[k]/i}, \begin{pmatrix}\mathbf{C}_0 \\ \mathbf{s}_i\mathbf{C}_0 + \mathbf{e}'\end{pmatrix}\right).$$

Using the leftover hash lemma with $\mathbf{A}_i$ as seed and $\mathbf{R}$ as source, we have $(\mathbf{A}_i, \mathbf{C}_0) \approx_s (\mathbf{A}_i, \mathbf{Z})$, where $\mathbf{Z} \leftarrow \frac{q}{d}\mathbb{Z}_d^{n\times m}$, thus

$$\left(\mathbf{A}_i, \{\mathbf{b}_{i,j}\}_{j\in[k]/i}, \begin{pmatrix}\mathbf{C}_0\\ \mathbf{s}_i\mathbf{C}_0+\mathbf{e}'\end{pmatrix}\right)$$
$$\approx_s \left(\mathbf{A}_i, \{\mathbf{b}_{i,j}\}_{j\in[k]/i}, \begin{pmatrix}\mathbf{Z}\\ \mathbf{s}_i\mathbf{Z}+\mathbf{e}'\end{pmatrix}\right).$$

We note that $\mathbf{Z}$ is independent of $\mathbf{s}_i$, as $\mathbf{C}_0$ is generated after $\mathbf{s}_i$ and $\mathbf{b}_{i,j}$. Assuming $\tilde{H}_\infty(\mathbf{s}_i|\{\mathbf{b}_{i,j}\}_{j\in[k]/i}) \geq n$, let $r = (n-\omega(\log n))/\log d$ and dual-Regev encryption is *circular security*. By Theorem 5, it holds that

$$\left(\mathbf{A}_i, \{\mathbf{b}_{i,j}\}_{j\in[k]/i}, \begin{pmatrix}\mathbf{Z}\\ \mathbf{s}_i\mathbf{Z}+\mathbf{e}'\end{pmatrix}\right) \approx_c \left(\mathbf{A}_i, \{\mathbf{b}_{i,j}\}_{j\in[k]/i}, \begin{pmatrix}\mathbf{Z}\\ \mathbf{z}\end{pmatrix}\right),$$

where $\mathbf{z} \leftarrow \mathbb{Z}_d^m$, Thus $(\mathbf{A}_i, \{\mathbf{b}_{i,j}\}_{j\in[k]/i}, \mathbf{C})$ and $(\mathbf{A}_i, \{\mathbf{b}_{i,j}\}_{j\in[k]/i}, \mathbf{U})$, are (jointly) computational indistinguishable. □

Note that the premise of the above result is that $\tilde{H}_\infty(\mathbf{s}_i|\{\mathbf{b}_{i,j}\}_{j\in[k]/i}) \geq n$ where $\mathbf{b}_{i,j} = \mathbf{s}_i\mathbf{A}_j + \mathbf{e}_j$. Without loss of generality, assuming $i = 1$ we have

$$(\mathbf{b}_{1,2}, \mathbf{b}_{1,3}, \cdots, \mathbf{b}_{1,k}) = \mathbf{s}_1(\mathbf{A}_2, \mathbf{A}_3, \cdots, \mathbf{A}_k) + (\mathbf{e}_2, \mathbf{e}_3, \cdots, \mathbf{e}_k).$$

Let $\bar{\mathbf{A}} = (\mathbf{A}_2, \mathbf{A}_3, \cdots, \mathbf{A}_k)$, $\bar{\mathbf{e}} = (\mathbf{e}_2, \mathbf{e}_3, \cdots, \mathbf{e}_k)$ by Theorem 2 if $0 < \sigma_1 < d/(2\sqrt{m(k-1)})$ we have

$$\tilde{H}_\infty(\mathbf{s}_i|\mathbf{s}_i\bar{\mathbf{A}} + \bar{\mathbf{e}}) \geq -\log\left(\frac{1}{\rho_{\sigma_1}(\Lambda_q(\bar{\mathbf{A}}))} + 2^{-m(k-1)}\right). \quad (7)$$

Let $\gamma > 0$ be real $\mathcal{B}$ be a sphere of radius 1, by Lemma 2 if $\mathsf{rank}(\Lambda_q(\bar{\mathbf{A}}) \bigcap \gamma\mathcal{B}) \geq n/2$ and $\sigma_1 > 5\gamma$, it holds that $\rho_{\sigma_1}(\Lambda_q(\bar{\mathbf{A}})) > 5^{n/2}$ (satisfying $1/\rho_{\sigma_1}(\Lambda_q(\bar{\mathbf{A}})) \leq 2^{-n} - 2^{-m(k-1)}$, thus $\tilde{H}_\infty(\mathbf{s}_i|\{\mathbf{b}_{i,j}\}_{j\in[k]/i}) \geq n$). We observe from Ananth et al. (2021) that one way to satisfy $\mathsf{rank}(\Lambda_q(\bar{\mathbf{A}}) \bigcap \gamma\mathcal{B}) \geq n/2$ is to make $\bar{\mathbf{A}}$ have structure as

$$\bar{\mathbf{A}} = \begin{pmatrix}\mathbf{B}_2 & \mathbf{B}_3 & \cdots & \mathbf{B}_k\\ \mathbf{S}\mathbf{B}_2+\mathbf{E}_2, & \mathbf{S}\mathbf{B}_3+\mathbf{E}_3, & \cdots, & \mathbf{S}\mathbf{B}_k+\mathbf{E}_k\end{pmatrix}.$$

where $\mathbf{B}_i \leftarrow \frac{q}{d}\mathbb{Z}_d^{\frac{n}{2}\times m}$, $\mathbf{S} \leftarrow \mathbb{Z}_d^{\frac{n}{2}\times\frac{n}{2}}$, $\mathbf{E}_i \leftarrow \mathcal{D}_{\frac{q}{d}\mathbb{Z}^{\frac{n}{2}\times m},\sigma_3}$ and $\sqrt{2n} < \sigma_3 < \gamma/\sqrt{m(k-1)}$. Thus it holds that

$$\begin{pmatrix}\mathbf{I} & \\ \mathbf{S} & \mathbf{I}\end{pmatrix}^{-1} \cdot \bar{\mathbf{A}} = \begin{pmatrix}\mathbf{B}_2, & \mathbf{B}_3, & \cdots, & \mathbf{B}_k\\ \mathbf{E}_2, & \mathbf{E}_3, & \cdots, & \mathbf{E}_k\end{pmatrix} \in \Lambda_q(\bar{\mathbf{A}}).$$

Let $\mathbf{V} = (\mathbf{E}_2, \mathbf{E}_3, \cdots, \mathbf{E}_k)$ be the lower half of the above matrix, it holds that $\mathbf{V} \in \Lambda_q(\bar{\mathbf{A}})$ and $\mathsf{rank}(\mathbf{V}) = n/2$. By Lemma 3 the length of all row vectors of $\mathbf{V}$ does not exceed γ, thus $\mathsf{rank}(\Lambda_q(\bar{\mathbf{A}}) \bigcap \gamma\mathcal{B}) \geq n/2$.

Put things together

We bring together the previous parameter requirements, in particular the range of standard deviations for several discrete Gaussian distributions. For (7) holds, we need $0 < \sigma_1 < d/(2\sqrt{m(k-1)})$. To make $\mathsf{rank}(\Lambda_q(\bar{\mathbf{A}}) \bigcap \gamma\mathcal{B}) \geq n/2$ and $\bar{\mathbf{A}}$ looks random, we need $\sigma_1 > 5\gamma$ and $\sqrt{2n} < \sigma_3 < \gamma/\sqrt{m(k-1)}$. In the $\mathsf{setup}(\cdot)$ of *Scheme*#2, we need $(m\sigma_1)/\sigma_2 = \mathsf{negl}(\lambda)$. To sum up, we get the standard deviation $\sigma_1, \sigma_2, \sigma_3$ respectively as follows

$$5\sqrt{2mn(k-1)} < \sigma_1 < d/(2\sqrt{m(k-1)}),$$
$$\sigma_2 = (m\sigma_1)/\mathsf{negl}(\lambda), \quad \sigma_3 \geq \sqrt{2n}.$$

Conclusion

In this paper, we investigated a more generalized leakage-resilient model in lattice-based cryptography and extended its application in decentralization. We focused on quantifying the Average Conditional Min-Entropy $\tilde{H}_\infty(\mathbf{s}|\mathbf{s}\mathbf{A}+\mathbf{e})$ for different secret distributions of $\mathbf{s}$, including uniform and discrete Gaussian distributions. Our results generalize the previous work on multi-key fully homomorphic encryption (MKFHE) and improve its efficiency by reducing the key and ciphertext sizes. By extending the active leakage model, we demonstrated that when the secret $\mathbf{s}$ follows a bounded uniform or discrete Gaussian distribution, it is still possible to derive similar entropy bounds as in previous studies. This allowed us to optimize the MKFHE scheme proposed in Brakerski et al. (2017) by transitioning from a Dual-GSW to a GSW-based ciphertext structure, leading to a more compact representation and efficient computations. Our improvements resulted in reduced ciphertext size, more efficient homomorphic multiplications, and lower communication costs in the setup phase. Further work will consider extending the above results to algebraic lattices, as well as the conclusions related to RLWE, to provide theoretical support for the decentralization of the MKFHE scheme based on RLWE.

Appendix

Averaging argument

For any random variable X, if the expectation of X is at least ρ, then there must be a value of X that is at least ρ. Namely

$$\mathsf{Exp}[X] \geq \rho \Longrightarrow \Pr[X \geq \rho] > 0.$$

see details: https://www.cs.princeton.edu/courses/archive/spr06/cos522/averaging.pdf.

Claim 3

If everyone likes at least $\frac{1}{3}$ of the books in the library, then there is a book in the library that at least $\frac{1}{3}$ of the people like.

Proof

Suppose the number of people and books are N, B, respectively. Ask everyone to mark their favorite book with a red dot. Thus, the red dot mark in the book in the library has at least $\frac{NB}{3}$. Now, assuming that there is no book that is liked by at least $\frac{1}{3}$ of the people, the number of red dot marks in each book would be less than $\frac{N}{3}$. This would result in the total number of red dot marks in the library being less than $\frac{NB}{3}$, which leads to a contradiction. □

The proof of Theorem 2

Proof

For a given $\mathbf{A} \in \frac{q}{d}\mathbb{Z}_d^{n\times m}$ and $\mathbf{y} \in \frac{q}{d}\mathbb{Z}_d^m$, let $\mathbf{s}^*$ be the point that maximizes the conditional probability $\Pr_{\mathbf{s}\leftarrow\mathbb{Z}_d^n}[\mathbf{s} = \mathbf{s}^*|\mathbf{y} = \mathbf{sA} + \mathbf{e}]$. By Bayes Rule, it holds that

$$\begin{aligned}\Pr_{\mathbf{s}\leftarrow\mathbb{Z}_d^n}[\mathbf{s} = \mathbf{s}^*|\mathbf{y} = \mathbf{sA} + \mathbf{e}] &= \Pr\left[\mathbf{y} = \mathbf{sA} + \mathbf{e} \mid \mathbf{s}^*\right] \cdot \frac{\Pr[\mathbf{s} = \mathbf{s}^*]}{\Pr[\mathbf{y} = \mathbf{sA} + \mathbf{e}]} \\ &= \frac{\Pr[\mathbf{e} = \mathbf{y} - \mathbf{s}^*\mathbf{A}]}{\sum_{\mathbf{s}'} \Pr[\mathbf{e} = \mathbf{y} - \mathbf{s}'\mathbf{A}]}.\end{aligned}$$

For the given $\mathbf{A}$ and $\mathbf{y}$, $\sum_{\mathbf{s}'} \Pr[\mathbf{e} = \mathbf{y} - \mathbf{s}'\mathbf{A}]$ is a constant, it holds that $\Pr[\mathbf{s} = \mathbf{s}^*|\mathbf{y} = \mathbf{sA} + \mathbf{e}] \propto \Pr[\mathbf{e} = \mathbf{y} - \mathbf{s}^*\mathbf{A}]$. Thus the point maximizes $\Pr[\mathbf{s} = \mathbf{s}^*|\mathbf{y} = \mathbf{sA} + \mathbf{e}]$ is the lattice point nearest to $\mathbf{y}$. Let $V \in \frac{q}{d}\mathbb{Z}^m$ be the *discretized Voronoïcell* of $\Lambda_q(\mathbf{A})$, that is V consists of all point in $\frac{q}{d}\mathbb{Z}^m$ that are closer to 0 than to any other point in Λ. By construction, V is a system of coset representatives of $\frac{q}{d}\mathbb{Z}^m \backslash \Lambda_q(\mathbf{A})$.

By Theorem 3, it holds that $\Pr[\mathbf{s} = \mathbf{s}^*|\mathbf{y} = \mathbf{sA} + \mathbf{e}] = \Pr[\mathbf{e} \mod q \in V]$. By Lemma 3, it holds that $||\mathbf{e}|| \leq \frac{q}{d}\sigma\sqrt{m} < q/2$ except with probability 2^{-m}, thus $\Pr[\mathbf{e} \mod q \in V] \leq \Pr[\mathbf{e} \in V] + 2^{-m}$. By Lemma 1, it holds that $\Pr[\mathbf{e} \in V] \leq \rho_\sigma(V)/\rho_\sigma(\frac{q}{d}\mathbb{Z}^m) \leq 1/\rho_\sigma(\Lambda_q(\mathbf{A}))$, therefore $\Pr[\mathbf{e} \mod q \in V] \leq 1/\rho_\sigma(\Lambda_q(\mathbf{A})) + 2^{-m}$. It holds that

$$\begin{aligned}\tilde{H}_\infty(\mathbf{s} \mid \mathbf{sA} + \mathbf{e}) &= -\log\left(\mathrm{E}_{\mathbf{y}}\left[\max_{\mathbf{s}^*} \Pr_{\mathbf{s},\mathbf{e}}[\mathbf{s} = \mathbf{s}^* \mid \mathbf{y} = \mathbf{sA} + \mathbf{e}]\right]\right) \\ &\geq -\log\left(\frac{1}{\rho_\sigma\left(\Lambda_q(\mathbf{A})\right)} + 2^{-m}\right).\end{aligned}$$

□

Regularity of hash functions on prime order groups and their Cartesian products

Before proving the regularity of hash functions mapped to prime order groups, we need to introduce Xagawa's version of the leftover hash lemma, which will be used in our proof.

Lemma 5

(*Lemma* 4.2.3 *in Xagawa* (2010)) *Let $\mathcal{H}_k = \{h_k : k \in K\}$ be a universal hash function defined over finite set K, D, T*

$$\begin{aligned} h_k : D &\to T \\ x &\mapsto h_k(x)\end{aligned}$$

where x is a random variable defined over D and independent from k. It holds that

$$\Delta((U, h_k(x)), (U, V)) \leq 2^{-\frac{1}{2}(\tilde{H}_\infty(x) - \log|T| + 2)},$$

where U and V are uniform random variable defined over K and T.

Let m be integer, $\mathbf{b} = (b_1, b_2, \cdots, b_m) \in \{0,1\}^m$, G be a finite Abelian group, $\mathbf{g} = (g_1, g_2, \cdots, g_m) \in G^m$. Regev (Claim 5.3 in Regev (2009)) proved that the hash function family $\{\mathcal{H}_G = h_{\mathbf{g}} : \mathbf{g} \in G^m\}$ is *universal* where

$$\begin{aligned} h_{\mathbf{g}} : \{0,1\}^m &\to G \\ \mathbf{b} &\mapsto \sum_{i\in m} b_i g_i\end{aligned}$$

The above result requires that the preimage of $h_{\mathbf{g}}$ is taken from $\mathbb{Z}_2$, and G only needs to be a finite Abelian group. Here we relax the preimage of $h_{\mathbf{g}}$ and take it from $\mathbb{Z}_d$ where $d < q$, but the order of the finite Abelian group G must be prime. We will prove that the following hash function family is *universal* $\mathcal{H}_G = \{h_{\mathbf{g}} : \mathbf{g} \in G^m\}$

$$h_{\mathbf{g}} : \mathbb{Z}_d^m \to G$$
$$\mathbf{b} \mapsto \sum_{i\in[m]} b_i g_i$$

we prove Lemma 6 and then extend it to its Cartesian products.

Lemma 6

Let G be a finite Abelian group with order $|G| = q$ as a prime, m, d as integers and $d \le q$. For uniformly chosen $g_1, \cdots, g_m \in G$, $b_i \leftarrow \mathbb{Z}_d$ and $u \leftarrow G$, the statistical distance $\Delta(\sum_{i\in[m]} b_i g_i, u)$ is expected to be at most $\frac{1}{2}\sqrt{\frac{q}{d^m}}$. In particular, the probability that the statistical distance exceeds $(\frac{q}{d^m})^{\frac{1}{4}}$ does not exceed $(\frac{q}{d^m})^{\frac{1}{4}}$.

Proof

As G is a finite Abelian group, it holds that for $\mathbf{b}, \mathbf{b}' \in \mathbb{Z}_d^m$, and $\mathbf{b} \neq \mathbf{b}'$, we have

$$\begin{aligned}&\Pr_{\mathbf{g}\leftarrow G^m}\left[\sum_i^m b_i g_i = \sum_{i=1} b_i' g_i \middle| \mathbf{b} \neq \mathbf{b}'\right] \\ &= \Pr_{\mathbf{g}\leftarrow G^m}\left[\sum_{i=1}^m (b_i - b_i') g_i = 0 \middle| \mathbf{b} \neq \mathbf{b}'\right].\end{aligned} \tag{8}$$

The above probability can be determined by counting the fixed $\mathbf{b}$ and $\mathbf{b}'$. When only the i-th element of $\mathbf{b} - \mathbf{b}'$ is non-zero, $b_i - b_i' \neq 0$, we have $\sum_{i=1}^m (b_i - b_i') g_i = (b_i - b_i') g_i = 0$. Because G is a finitely generated Abelian group with prime order q, then G and $\mathbb{Z}_q$ are isomorphic. For any $b_i, b_i' \in \mathbb{Z}_d$, and $b_i - b_i' \neq 0$, there is $b_i - b_i' \in [-(d-1), d-1]/0$. Therefore, for $(b_i - b_i') g_i = 0$, we have $g_i = 0$, and the remaining $m-1$ positions can be chosen randomly on G, thus $(8) = q^{m-1}/q^m = 1/q$.

When only the i-th and j-th elements of $\mathbf{b} - \mathbf{b}'$ are non-zero, we have $(b_i - b_i') g_i + (b_j - b_j') g_j = 0$ then $g_i = -(b_i - b_i')^{-1}(b_j - b_j') g_j$ where $(b_i - b_i')^{-1}$ is the inverse of $b_i - b_i'$ on $\mathbb{Z}_q$. For a given g_j, g_i is uniquely determined, and the remaining $m-2$ positions can be arbitrarily selected on G, thus $(8) = q^{m-2} q/q^m = 1/q$. Generally, when only k elements of $\mathbf{b} - \mathbf{b}'$ are non-zero, it can be derived from the linear relationship

$$g_i = -(b_i - b_i')^{-1} \sum_{j\in[k]/i} (b_j - b_j') g_j,$$

it holds that

$$\Pr_{\mathbf{g}\leftarrow G^m}\left[\sum_i^m b_i g_i = \sum_{i=1}^m b_i' g_i \middle| \mathbf{b} \neq \mathbf{b}'\right] = \frac{q^{m-k} q^{k-1}}{q^m} = 1/q.$$

Therefore, the family of hash functions $\mathcal{H}_G = \{h_{\mathbf{g}} : \mathbf{g} \in G^m\}$ defined above are *universal*. In particular, for$\mathbf{b} \leftarrow \mathbb{Z}_d^m$ the probability of collision is $1/d^m$ the min-entropy is $m \log d$ and the output of this hash function is $\log q$ bits. Let $\epsilon = \sqrt{2}^{(\log q - m \log d)}$. By Lemma 5 it holds that

$$\Delta((\mathbf{g}, \sum_i b_i g_i), (\mathbf{g}, u)) \le \frac{1}{2}\epsilon \le \sqrt{\frac{q}{d^m}},$$

where $u \leftarrow G$. The following estimate of the statistical distance expectation is similar to Lemma 4.3.3 in Xagawa (2010). For any $\mathbf{g} = (g_1, \cdots, g_m) \in G^m$ define

$$P_{\mathbf{g}}(h) = \frac{1}{d^m}\left|\left\{\mathbf{b} \in \mathbb{Z}_d^m : \sum_{i=1}^m b_i g_i = h\right\}\right|.$$

For a fixed $\mathbf{g} \in G^m$ define the collision boundary, that is the l_2-norm of the function $P_{\mathbf{g}}$ on $\mathbb{R}^q$

$$\begin{aligned}\sum_{h\in G} P_{\mathbf{g}}(h)^2 &= \Pr_{\mathbf{b},\mathbf{b}'\leftarrow\mathbb{Z}_d^m}\left[\sum_{i=1}^m b_i g_i = \sum_{i=1}^m b_i' g_i\right] \\ &\le \frac{1}{d^m} + \Pr_{\mathbf{b},\mathbf{b}'\leftarrow\mathbb{Z}_d^m}\left[\sum_{i=1}^m b_i g_i = \sum_{i=1}^m b_i' g_i \middle| \mathbf{b} \neq \mathbf{b}'\right].\end{aligned}$$

Thus for random variable $\mathbf{g}$, it hold that

$$\mathsf{Exp}_{\mathbf{g}\leftarrow G^m}\left[\sum_{h\in G} P_{\mathbf{g}}(h)^2\right] \in \frac{1}{d^m} \pm \frac{1}{q}.$$

For any $\mathbf{x} \in \mathbb{R}^q$ it holds that $||\mathbf{x}||_\infty \le \sqrt{q}||\mathbf{x}||_2$ we have

$$\begin{aligned}\mathsf{Exp}_{\mathbf{g}\leftarrow G^m}\left[\sum_{h\in\mathbf{G}}\left|P_{\mathbf{g}}(h) - \frac{1}{q}\right|\right] &\le \mathsf{Exp}_{\mathbf{g}\leftarrow G^m}\left[q^{1/2}\left(\sum_{h\in\mathbf{G}}\left(P_{\mathbf{g}}(h) - \frac{1}{q}\right)^2\right)^{1/2}\right] \\ &= q^{1/2}\mathsf{Exp}_{\mathbf{g}\leftarrow G^m}\left[\left(\sum_{h\in\mathbf{G}}\left(P_{\mathbf{g}}(h) - \frac{1}{q}\right)^2\right)^{1/2}\right] \\ &\le q^{1/2}\left(\mathsf{Exp}_{g\leftarrow G^m}\left[\sum_{h\in\mathbf{G}} P_{\mathbf{g}}(h)^2\right] - \frac{1}{q}\right)^{1/2} \\ &\le q^{1/2}\cdot d^{-m/2} = \sqrt{\frac{q}{d^m}}.\end{aligned}$$

Thus

$$\mathsf{Exp}_{\mathbf{g}\leftarrow G^m}\left[\Delta\left(\sum_i b_i g_i, u\right)\right] \leq \frac{1}{2}\sqrt{\frac{q}{d^m}}.$$

By the *averaging argument* (See Appendix A), we have

$$\Pr_{\mathbf{g}\leftarrow G^m}\left[\Delta\left(\sum_i b_i g_i, u\right) \geq \left(\frac{q}{d^m}\right)^{\frac{1}{4}}\right] \leq \left(\frac{q}{d^m}\right)^{\frac{1}{4}}.$$

(Otherwise, it can be derived that $\mathsf{Exp}_{\mathbf{g}\leftarrow G^m}\left[\Delta\left(\sum_i b_i g_i, u\right)\right] \geq \frac{1}{2}\sqrt{\frac{q}{d^m}}$, contradictory) □

Next, we extend the above lemma to the Cartesian product of prime order groups.

Corollary 1

Let k, m be an integers, $G_1 \times G_2 \cdots \times G_k$ be the Cartesian product of finite Abelian groups $\{G_i\}_{i\in[k]}$ where $|G_i| = q_i$ be primes, $q_{\min} = \min\{q_i\}_{i\in[k]}$. For the uniformly chosen $\{\mathbf{g}_t\}_{t\in[m]}$ where $\{\mathbf{g}_t = (g_{t,1}, \cdots g_{t,k}) \in G_1 \times G_2 \cdots \times G_k\}_{t\in[m]}$, $b_i \leftarrow \mathbb{Z}_{q_{\min}}$, $\mathbf{u} \leftarrow (G_1 \times G_2 \cdots \times G_k)$, the statistical distance $\Delta((\sum_{t=1}^m b_t g_{t,1}, \cdots, \sum_{t=1}^m b_t g_{t,k}), \mathbf{u})$ is expected to be at most $(\prod_{i=1}^k q_i/q_{\min}^m)^{\frac{1}{2}}$, in particular, the probability that the statistical distance exceeds $(\prod_{i=1}^k q_i/q_{\min}^m)^{\frac{1}{4}}$ does not exceed $(\prod_{i=1}^k q_i/q_{\min}^m)^{\frac{1}{4}}$.

Proof

Similar to the Lemma 6, we first prove the family of hash functions $\mathcal{H}_{G_1\times G_2,\cdots,G_k} = \{h_{\mathbf{g}} : \mathbf{g} \in G_1 \times G_2, \cdots, G_k\}$ is universal

$$h_{\mathbf{g}} : \mathbb{Z}_{q_{\min}}^m \to G_1 \times \cdots \times G_k$$
$$\mathbf{b} \mapsto \left(\sum_{t=1}^m b_i g_{i,1}, \sum_{t=1}^m b_i g_{i,2}, \cdots, \sum_{t=1}^m b_i g_{i,k}\right).$$

When $\mathbf{b} \neq \mathbf{b}' \mod q_{\min}$, and only the α-th element is non-zero $\{(b_\alpha - b'_\alpha)g_{\alpha,j} = 0\}_{j\in[k]}$, it holds that $\{g_{\alpha,j} = 0\}_{j\in[k]}$, then the collision probability is $\frac{1}{\prod_{i=1}^k q_i}$. Similarly, only when the α-th and β-th elements are non-zero $\{(b_\alpha - b'_\alpha)g_{\alpha,j} + (b_\beta - b'_\beta)g_{\beta,j} = 0\}_{j\in[k]}$, the collision probability is $\frac{1}{\prod_{i=1}^k q_i}$. Generally, when t elements are non-zero, it holds that

$$\left\{\sum_{\substack{i\in S\\ S\subset[m]}}^{|S|=t} (b_i - b'_i)g_{i,j} = 0\right\}_{j\in[k]},$$

the collision probability is $\frac{1}{\prod_{i=1}^k q_i}$. Thus $\mathcal{H}_{G_1\times G_2,\cdots,G_k}$ is *universal*. The proof of the statistical distance is similar to the Lemma 6, which will not be repeated here. □

Remark

We want to extend the above result to the general finite Abelian group, but the hash function mapping to it seems not to be *universal* (there is zero divisor). Such as, let $G \simeq \mathbb{Z}_q \times \mathbb{Z}_a$ (q be prime, $a > q$ be an integer), $b \neq b' \mod q$. For any $g_1 \leftarrow \mathbb{Z}_q$, $g_2 \leftarrow \mathbb{Z}_a$, let $(b - b')g_1 = 0$, it holds that $g_1 = 0$, but $(b - b')g_2 = 0$ holds for any $b - b'$ satisfying $ord(g_2)|(b - b')$, where $order(g_2)$ is the order of g_2, which the probability of $(b - b')g_2 = 0$ is

$$\Pr_{\substack{b,b'\leftarrow\mathbb{Z}_q\\ g_2\leftarrow G}}[(b - b')g_2 = 0|b - b' \neq 0]$$
$$= \sum_{i=1}^{q-1}\left(\Pr_{b,b'\leftarrow\mathbb{Z}_q}[(b - b') = i] \cdot \frac{gcd(i, a)}{a}\right).$$

The above probability is clearly greater than $\frac{1}{a}$.

Leakage-resistant properties of LWE over $\frac{q}{d}\mathbb{Z}^m$

In this section we introduce the LWE problem on $\frac{q}{d}\mathbb{Z}^m$, and then prove its leakage-resilient property. In fact, it will not be simpler than the standard LWE problem. We introduce this non-standard LWE problem, then reduce it to the standard LWE problem (which is almost an observation), and finally prove its leakage-resilient property. The following definitions and results will be used in the construction and security proof of our *Scheme*#2.

The DLWE problem over $\frac{q}{d}\mathbb{Z}^m$

Definition 3 For $n, m, d, q \in \mathbb{N}$ and $d \leq q$, the rational-$\mathsf{DLWE}_{n,m,d,q,\sigma}$ problem is to distinguish the following distribution

- $\mathcal{D}_0$: the joint distribution $(\mathbf{A}, \mathbf{z}) \in (\frac{q}{d}\mathbb{Z}_d^{n\times m} \times \frac{q}{d}\mathbb{Z}_d^m)$ is sampled by $\mathbf{A} \leftarrow \frac{q}{d}\mathbb{Z}_d^{n\times m}$, $\mathbf{z} \leftarrow \frac{q}{d}\mathbb{Z}_d^m$.
- $\mathcal{D}_1$: the joint distribution $(\mathbf{A}, \mathbf{b}) \in (\frac{q}{d}\mathbb{Z}_d^{n\times m} \times \frac{q}{d}\mathbb{Z}_d^m)$ is computed by $\mathbf{A} \leftarrow \frac{q}{d}\mathbb{Z}_d^{n\times m}$, $\mathbf{b} = \mathbf{sA} + \mathbf{e}$, where $\mathbf{s} \leftarrow \mathbb{Z}_d^n$, $\mathbf{e} \leftarrow \mathcal{D}_{\frac{q}{d}\mathbb{Z}^m,\sigma}$.

As introduced in Preliminary, the standard $\mathsf{DLWE}_{n,m,q,\sigma}$ is defined on $\mathbb{Z}^m$ with a standard deviation $\sigma > 2\sqrt{n}$ which not be simpler than the hard problem on

lattice. Now we build the reduction from $\mathsf{DLWE}_{n,m,q,\sigma}$ to $\mathsf{rational\text{-}DLWE}_{n,m,d,q,\sigma}$.

Claim 4

If there is an adversary can distinguish the $\mathsf{rational\text{-}DLWE}_{n,m,d,q,\sigma}$ *problem with an advantage of* ϵ *in time* T*, then he can also distinguish the standard* $\mathsf{DLWE}_{n,m,q,\sigma}$ *problem with the same time and advantage.*

Proof

In Sect. "The leakage-resilient of LWE samples with bounded uniform secrets and Gaussian noise", we have shown that there is a bijection between $\Lambda_d(\mathbf{A})$ *and* $\Lambda_q(\mathbf{A}')$ *where* $\mathbf{A}' = \frac{q}{d}\mathbf{A}, \mathbf{A} \in \mathbb{Z}_d^{n\times m}$*. Similarly, there is a bijection between* $\mathsf{DLWE}_{n,m,q,\sigma}$ *and* $\mathsf{rational\text{-}DLWE}_{n,m,d,q,\sigma}$ *samples. For any given standard* $\mathsf{DLWE}_{n,m,d,\sigma}$ *samples* $(\mathbf{A}, \mathbf{b} = \mathbf{sA} + \mathbf{e} \mod d)$*, let* $\mathbf{b} = \mathbf{sA} + \mathbf{e} + d \cdot \mathbf{c}$ *where* $\mathbf{c} \in \mathbb{Z}^m$*, it holds that*

$$\frac{q}{d}\mathbf{b} = \frac{q}{d} \cdot \mathbf{sA} + \frac{q}{d}\mathbf{e} + q \cdot \mathbf{c}.$$

Let $\mathbf{A}' = \frac{q}{d}\mathbf{A}, \mathbf{b}' = \frac{q}{d}\mathbf{b}, \mathbf{e}' = \frac{q}{d}\mathbf{e}$, it holds that

$$\mathbf{b}' = \mathbf{sA}' + \mathbf{e}' \mod q,$$

where $\mathbf{A}' \in \frac{q}{d}\mathbb{Z}_d^{n\times m}$, $\mathbf{e}' \in \frac{q}{d}\mathbb{Z}^m$. Thus $(\mathbf{A}', \mathbf{b}')$ are samples of $\mathsf{rational\text{-}DLWE}_{n,m,d,q,\sigma}$. For the standard deviation $\sigma > 2\sqrt{n}$, it will not be simpler than $\mathsf{DLWE}_{n,m,q,\sigma}$. □

Leakage resistance of rational LWE samples

Goldwasser et al. (2010) proved the leakage-resilient property of such "weak" LWE samples $(\mathbf{A}, \mathbf{b} = \mathbf{sA} + \mathbf{e})$ where $\mathbf{s}$ is taken from $\{0,1\}^n$ (as the entropy of $\mathbf{s}$ is sufficient, it is no simpler than the low-dimensional standard LWE problem). It essentially utilizes the leakage-resilient property of the leftover hash lemma and reduces it to low-dimensional LWE samples.

Next, we will prove that the $\mathsf{rational\text{-}DLWE}_{n,m,d,q,\sigma}$ samples also possess leakage-resilient properties. This proof needs to utilize the regularity result of the hash function family on the prime order group (Corollary 1), and unlike Shafi et al. (2010), we need to use the *circular security* assumption.

Theorem 5 *Let* n, q *be integers and* $d \le q$ *be prime,* $\mathbf{s}$ *be a random variable over* $\mathbb{Z}_d^n$*, having min-entropy at least* k*. For any* $r \le \frac{k-\omega(\log n)+2}{\log d}$*, there is a ppt reduction from* $\mathsf{rational\text{-}DLWE}_{n,m,d,q,\sigma}$ *to distinguish dual Regev ciphertext (with public key* $\mathsf{pk} = (\mathbf{B}, \mathbf{t})$*,* $\mathbf{B} \leftarrow \frac{q}{d}\mathbb{Z}_d^{n\times r}, \mathbf{t} = \mathbf{sB} \mod q$*, and plaintext is related to private key* $\mathbf{s}$*) defined over* $\frac{q}{d}\mathbb{Z}$ *from uniform distribution.*

Proof

For the given $\mathsf{rational\text{-}DLWE}$ *samples* $(\mathbf{A}, \mathbf{b} = \mathbf{sA} + \mathbf{e})$ *where* $\mathbf{A} \leftarrow \frac{q}{d}\mathbb{Z}^{n\times m}, \mathbf{e} \leftarrow \mathcal{D}_{\frac{q}{d}\mathbb{Z}^m,\sigma}, \mathbf{s} \leftarrow \mathbb{Z}_d^n$ *and* $\tilde{H}_\infty(\mathbf{s}) \ge k$*, by Claim 4 we can replace* $\mathbf{A}$ *with* $\mathbf{BC} + \mathbf{E}$*, where* $\mathbf{B} \leftarrow \frac{q}{d}\mathbb{Z}_d^{n\times r}, \mathbf{C} \leftarrow \mathbb{Z}_d^{r\times m}, \mathbf{E} \leftarrow \mathcal{D}_{\frac{q}{d}\mathbb{Z}^{n\times m},\sigma}$*. It holds that* $\mathbf{b} = \mathbf{sBC} + \mathbf{sE} + \mathbf{e}$*. As* $\frac{q}{d}\mathbb{Z}_d$ *is a finite Abelian group with* d *elements, for any randomly chosen* $\mathbf{B} \leftarrow \frac{q}{d}\mathbb{Z}_d^{n\times r}$*,* $h_{\mathbf{B}}$ *defines a hash function* $\mathcal{H}_{\frac{q}{d}\mathbb{Z}_d^r} = \{h_{\mathbf{B}} : \mathbf{B} \in \frac{q}{d}\mathbb{Z}_d^{n\times r}\}$ *mapping from* $\mathbb{Z}_d^n$ *to* $\frac{q}{d}\mathbb{Z}_d^r$*.*

$$h_{\mathbf{B}} : \mathbb{Z}_d^n \to \frac{q}{d}\mathbb{Z}_d^r$$
$$\mathbf{s} \mapsto \mathbf{sB} \mod q.$$

By Corollary 1 the family of hash functions $\mathcal{H}_{\frac{q}{d}\mathbb{Z}_d^r}$ is *universal*. Furthermore, by Lemma 5 we have

$$\Delta((\mathbf{B}, \mathbf{t}), (\mathbf{B}, \mathbf{u})) \le 2^{-\frac{1}{2}(\tilde{H}_\infty(\mathbf{s}) - \log d^r + 2)}.$$

For any $r \le \frac{k-\omega(\log n)+2}{\log d}$, we have $\Delta((\mathbf{B}, \mathbf{t}), (\mathbf{B}, \mathbf{u})) \le \mathsf{negl}(n)$ where $\mathbf{u} \leftarrow \frac{q}{d}\mathbb{Z}_d^r$. We can rewritten $\mathbf{b} = \mathbf{tC} + \mathbf{sE} + \mathbf{e}$, note that $\mathbf{tC} + \mathbf{e}$ are m rational LWE samples. In Shafi et al. (2010), they set the variance of $\mathbf{E}$ and $\mathbf{e}$ to satisfy $||\mathbf{sE}||/||\mathbf{e}|| = \mathsf{negl}(n)$, then get $\mathbf{e} \approx_s \mathbf{e} + \mathbf{sE}$. Thus $\mathbf{b} \approx_{\mathbf{s}} \mathbf{tC} + \mathbf{e}$ and the dimensional-r standard LWE samples are indistinguishable. However, this method is not suitable for us as $\mathcal{D}_{\frac{q}{d}\mathbb{Z}^{n\times m},\sigma}$ is defined on $\frac{q}{d}\mathbb{Z}^{n\times m}$ and $\mathbf{s} \leftarrow \mathbb{Z}_d^n$, the norm of $\mathbf{sE}$ will overturn q with a high probability, which leaves no room for us to set the variance of $\mathbf{E}$ and $\mathbf{e}$ to satisfy $||\mathbf{sE}||/||\mathbf{e}|| = \mathsf{negl}(n)$.

We noticed that $\mathbf{b} = \mathbf{tC} + \mathbf{e} + \mathbf{sE}$ could be regarded as the ciphertext of the dual Regev encryption, where $(\mathbf{B}, \mathbf{t} = \mathbf{sB})$ is the public key, $\mathbf{s}$ is the private key, and $\mathbf{sE}$ is the plaintext (related to the private key). Suppose we assume the dual-Regev encryption scheme is *circular security* (while the encrypted data is related to private key, the ciphertext is still computationally indistinguishable). In that case, we should have enough confidence in the leak resistance of $\mathsf{rational\text{-}DLWE}_{n,m,d,q,\sigma}$.

Considering that the *circular security* assumption exists in many places, such as the *key-switch* in the FHE scheme (Brakerski et al. 2012) and the bootstrapping in Gentry (2009); Brakerski and Vaikuntanathan (2011); Gentry et al. (2013). Therefore, if an adversary can distinguish the $\mathsf{rational\text{-}DLWE}_{n,m,d,q,\sigma}$ with the private key $\mathbf{s}$ lossy,

then he can distinguish the dual Regev ciphertext (with plaintext is related to the private key). □

Acknowledgements
The authors would like to thank anonymous reviewers for their helpful comments and suggestions.

Author contributions
Xiaokang Dai: Conceptualization, Methodology, Writing—Original Draft. Jingwei Chen: Conceptualization, Methodology, Writing—Original Draft. Wenyuan Wu: Supervision, Writing—Review. Yong Feng: Editing, Funding Acquisition—Review. All authors read and approved the final manuscript.

Funding
This study was funded by National Key Research and Development Project (2020YFA0712300), National Natural Science Foundation of China (12571553), Academicians in Chongqing Lead the Scientific and Technological Innovation Guidance Project (CSTB2023YSZX-JCX0008), Chongqing Talents Plan Youth Top-Notch Project (2021000263), Western Young Scholars Program of CAS.

Data availability
Not applicable.

Declarations

Conflict of interest
The authors declare that they have no Conflict of interest.

Received: 3 March 2025 Accepted: 20 June 2025
Published online: 26 November 2025

References

Alperin-Sheriff J, Peikert C (2014) Faster bootstrapping with polynomial error. In: Garay JA, Gennaro R (eds) Advances in cryptology—CRYPTO 2014. Springer, Berlin Heidelberg, Berlin, Heidelberg, pp 297–314
Ananth P, Jain A, Jin Z, Malavolta G (2020) Multi-key fully-homomorphic encryption in the plain model. In: Pass R, Pietrzak K (eds) Theory of cryptography. Springer International Publishing, Cham, pp 28–57
Ananth P, Jain A, Jin Z, Malavolta G (2021) Unbounded multi-party computation from learning with errors. In: Canteaut A, Standaert FX (eds) Advances in cryptology—EUROCRYPT 2021. Springer International Publishing, Cham, pp 754–781
Applebaum B, Cash D, Peikert C, Sahai A (2009) Fast cryptographic primitives and circular-secure encryption based on hard learning problems. In: Halevi S (ed) Advances in cryptology—CRYPTO 2009. Springer, Berlin Heidelberg, Berlin, Heidelberg, pp 595–618
Asharov G, Jain A, López-Alt A, Tromer E, Vaikuntanathan V, Wichs D (2012) Multiparty computation with low communication, computation and interaction via threshold FHE. In: Pointcheval D, Johansson T (eds) Advances in cryptology—EUROCRYPT 2012. Springer, Berlin Heidelberg, Berlin, Heidelberg, pp 483–501
Banaszczyk W (1993) New bounds in some transference theorems in the geometry of numbers. Math Ann 296:625–635
Brakerski Z, Döttling N (2018) Two-message statistically sender-private OT from LWE. In: Beimel A, Dziembowski S (eds) Theory of cryptography. Springer, Cham, pp 370–390
Brakerski Z, Döttling N (2020) Hardness of LWE on general entropic distributions. In: Canteaut A, Ishai Y (eds) Advances in cryptology—EUROCRYPT 2020. Springer, Cham, pp 551–575
Brakerski Z, Döttling N (2020) Lossiness and entropic hardness for ring-LWE. In: Pass R, Pietrzak K (eds) Theory of cryptography. Springer, Cham, pp 1–27
Brakerski Z, Gentry C, Vaikuntanathan V (2012) (leveled) fully homomorphic encryption without bootstrapping. In: Proceedings of the 3rd innovations in theoretical computer science conference. ITCS '12, Association for Computing Machinery, New York, NY, USA, pp. 309–325. https://doi.org/10.1145/2090236.2090262
Brakerski Z, Halevi S, Polychroniadou A (2017) Four round secure computation without setup. In: Kalai Y, Reyzin L (eds) Theory of cryptography. Springer, Cham, pp 645–677
Brakerski Z, Vaikuntanathan V (2011) Efficient fully homomorphic encryption from (standard) lwe. In: Proceedings of the 2011 IEEE 52nd annual symposium on foundations of computer science. FOCS '11, IEEE Computer Society, USA, pp 97–106. https://doi.org/10.1109/FOCS.2011.12
Chen H, Dai W, Kim M, Song Y (2019) Efficient multi-key homomorphic encryption with packed ciphertexts with application to oblivious neural network inference. In: Proceedings of the 2019 ACM SIGSAC conference on computer and communications security. CCS '19, Association for Computing Machinery, New York, NY, USA, pp 395–412. https://doi.org/10.1145/3319535.3363207
Chen L, Zhang Z, Wang X (2017) Batched multi-hop multi-key FHE from ring-LWE with compact ciphertext extension. In: Kalai Y, Reyzin L (eds) Theory of cryptography. Springer, Cham, pp 597–627
Chillotti I, Gama N, Georgieva M, Izabachène M (2016) Faster fully homomorphic encryption: Bootstrapping in less than 0.1 seconds. In: Cheon JH, Takagi T (eds) Advances in cryptology–ASIACRYPT 2016. Springer Berlin Heidelberg, Berlin, Heidelberg. pp. 3–33
Clear M, McGoldrick C (2015) Multi-identity and multi-key leveled FHE from learning with errors. In: Gennaro R, Robshaw M (eds) Advances in cryptology—CRYPTO 2015. Springer, Berlin, pp 630–656
Dai X, Chen J, Wu W, Feng Y (2024) Lattice-based, more general anti-leakage model and its application in decentralization. In: Zhu T, Li Y (eds) Information security and privacy. Springer, Singapore, pp 44–63
Dai X, Wu W, Feng Y (2022) Key lifting : Multi-key fully homomorphic encryption in plain model without noise flooding. Cryptology ePrint Archive, Paper 2022/055 , https://eprint.iacr.org/2022/055
Ducas L, Micciancio D (2015) Fhew: bootstrapping homomorphic encryption in less than a second. In: Oswald E, Fischlin M (eds) Advances in cryptology—EUROCRYPT 2015. Springer, Berlin, pp 617–640
Gentry C (2009) Fully homomorphic encryption using ideal lattices. In: Proceedings of the forty-first annual ACM symposium on theory of computing. STOC '09, Association for Computing Machinery, New York, NY, USA, pp 169–178. https://doi.org/10.1145/1536414.1536440
Gentry C, Sahai A, Waters B (2013) Homomorphic encryption from learning with errors: conceptually-simpler, asymptotically-faster, attribute-based. In: Canetti R, Garay JA (eds) Advances in cryptology—CRYPTO 2013. Springer, Berlin, pp 75–92
Jin H, Liu FH, Wang Z, Gu D (2024) Discrete gaussians modulo sub-lattices: New leftover hash lemmas for discrete gaussians. Cryptology ePrint Archive, Paper 2024/1695, https://eprint.iacr.org/2024/1695
López-Alt A, Tromer E, Vaikuntanathan V (2012) On-the-fly multiparty computation on the cloud via multikey fully homomorphic encryption. In: Proceedings of the forty-fourth annual ACM symposium on theory of computing. STOC '12, Association for Computing Machinery, New York, NY, USA, pp. 1219–1234. https://doi.org/10.1145/2213977.2214086
Micciancio D, Peikert C (2012) Trapdoors for lattices: simpler, tighter, faster, smaller. In: Pointcheval D, Johansson T (eds) Advances in cryptology—EUROCRYPT 2012. Springer, Berlin, pp 700–718
Mukherjee P, Wichs D (2016) Two round multiparty computation via multi-key FHE. In: Fischlin M, Coron JS (eds) Advances in cryptology—EUROCRYPT 2016. Springer, Berlin, pp 735–763
Peikert C, Shiehian S (2016) Multi-key FHE from LWE, revisited. In: Theory of cryptography conference. Springer. pp. 217–238
Regev O (2009) On lattices, learning with errors, random linear codes, and cryptography. J ACM. https://doi.org/10.1145/1568318.1568324
Shafi G, Yael K, Chris P, Vinod V (2010) Robustness of the learning with errors assumption. In: Gennaro R, Robshaw M (eds) Innovations in Computer Science. pp. 230–240
Xagawa K (2010) Cryptography with lattices. Ph.D. thesis, Department of Mathematical and Computing Sciences Tokyo Institute of Technology. Available at http://xagawa.net/pdf/2010Thesis.pdf
Xu J, Wu W, Feng Y, Dong R (2025) Monotonic optimization with application to the selection of parameters for LWE-based encryption schemes. J Syst Sci Complex
Yao AC (1982) Protocols for secure computations. In: 23rd Annual symposium on foundations of computer science (sfcs 1982). pp. 160–164

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Wenyuan Wu professor, director of Automatic Reasoning and Cognition Center of CIGIT. His research interests include automatic reasoning, cryptography, privacy computing, etc. He has published more than 50 papers in journal or conferences including Found. Comput. Math., Theor. Comput. Sci., etc., and was selected into the Light of the West program of CAS and Innovation Leaders Program of Chongqing Talents. He serves in the Computer Mathematics Committee of the Chinese Mathematical Society and in the editorial board of journal JSSC. He also serves as a reviewer for the Mathematical Reviews of the American Mathematical Society. http://english.cigit.cas.cn/sourcedb/ywrc/202311/t20231105_565222.html for more information.